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Research article

Spatiotemporal response of urban bike-sharing ridership to weather, air quality, and future climate change in major U.S. cities

Jangho Lee [a,b,*], Max Berkelhammer [a]

[a] Earth and Environmental Sciences, University of Illinois Chicago, Chicago, IL, 60607, USA
[b] Courant Institute School of Mathematics, Computing, and Data Science, New York University, New York, NY, 10012, USA

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ABSTRACT

Bike-sharing systems have become an essential component of sustainable urban transport, yet the resilience of system usage to changing environmental conditions remains insufficiently understood. This study provides a comprehensive spatiotemporal assessment of how weather and air quality influence bike-sharing ridership across five major U.S. cities between 2020 and 2024, and how these dynamics may evolve under future climate scenarios. Using generalized additive models, we reveal that primary weather variables mathematically dominate cycling decisions: ridership peaks around 20–25°C, while precipitation consistently suppresses usage. Conversely, air quality exerts a much weaker, secondary influence characterized by a behavioral dichotomy. Invisible, routine pollutants like ozone act as spurious proxies for pleasant weather, whereas physically perceptible hazards—such as acute wildfire smoke in San Francisco—can trigger sharp declines in usage. Hot spot analyses further show that environmental stress dynamically reconfigures spatial activity, driving a "climatic refuge" effect where cycling shifts toward waterfronts during extreme heat. Projecting these sensitivities forward, we show that future warming will enhance annual cycling suitability, particularly in seasonally cold cities, by reducing prohibitive winter days. Collectively, these results provide an integrated framework for understanding micromobility resilience, highlighting that urban cyclists respond primarily to immediate sensory environments rather than abstract health metrics.

1. Introduction

1.1. Bike sharing as sustainable urban mobility

Urban transportation and mobility systems are currently undergoing a profound, rapid, and necessary transformation. As global populations increasingly concentrate within metropolitan regions, cities are forced to confront the compounding crises of escalating greenhouse gas emissions, deteriorating localized air quality, and chronic, economically debilitating traffic congestion (Ding et al., 2025; Gifford and Steg, 2005; Mo et al., 2018; Wang and Zhou, 2017). In response to these multifaceted urban challenges, urban planners and policymakers have prioritized the transition toward low-carbon, active, and shared transportation modalities (Winkler et al., 2023). Within this shifting paradigm, shared micromobility, most notably docked and dockless bike-sharing systems, has emerged as a foundational component of the modern sustainable urban transport network.

Bike-sharing systems have evolved into critical civic infrastructure that provides flexible and affordable mobility options to millions of urban residents (Fishman, 2016; Shaheen et al., 2010). The rapid expansion and adoption of these systems contributes directly to broad climate change mitigation efforts by decarbonizing passenger kilometers traveled (Chapman, 2007; Creutzig and He, 2009). Beyond global climate targets, the localized integration of high-density bike-sharing networks can be linked to the reduction of surface urban heat island intensities through the displacement of combustion engine-generated anthropogenic heat (Lee and Berkelhammer, 2025). Furthermore, the reduction in vehicular exhaust inherently improves localized air quality, while the promotion of active travel yields cascading public health benefits, including enhanced cardiovascular health, lowered blood pressure among habitual users, and fundamentally improved urban livability (Agarwal et al., 2024; Guo et al., 2022).

However, despite these overwhelming socio-environmental benefits, the resilience of bike-sharing systems remains a subject of intense operational scrutiny. Unlike enclosed forms of mass transit or private automobiles, cycling requires direct, unmitigated physical exposure to the ambient atmospheric environment. Consequently, ridership demand

* Corresponding author. Earth and Environmental Sciences, University of Illinois Chicago, Chicago, IL, 60607, USA.
E-mail address: jangho.lee@nyu.edu (J. Lee).

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研究论文

美国主要城市城市共享单车使用量对天气、空气质量及未来气候变化的时空响应

Jangho Lee [a,b,*], Max Berkelhammer [a]

[a] 伊利诺伊大学芝加哥分校地球与环境科学系，美国伊利诺伊州芝加哥市，邮编 60607
[b] 纽约大学库朗研究所数学、计算与数据科学学院，美国纽约州纽约市，邮编 10012

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ABSTRACT

共享单车系统已成为可持续城市交通体系中的关键组成部分，然而该系统使用量对气候变化条件的韧性仍尚未被充分认知。本研究针对2020年至2024年间美国五大城市，系统性地开展了时空分析，探究天气与空气质量对共享单车使用量的影响，以及这些动态关系在未来气候情景下可能如何演变。通过采用广义加性模型，我们发现主要天气变量在骑行决策中具有数学主导作用：使用量在20–25 °C时达到峰值，而降水则持续抑制使用量；相反，空气质量的影响则显著较弱，且呈现出行为层面的双重效应——诸如臭氧等无形常规污染物仅作为宜人天气的虚假替代指标，而可感知的物理性危害（如旧金山地区的突发野火烟雾）则可能导致使用量骤降。热点分析进一步表明，环境压力会动态重构空间活动格局，进而催生“气候避难所”效应：在极端高温天气下，骑行活动会向滨水区域转移。基于上述敏感性分析进行前瞻性推演，我们发现未来气候变暖将通过减少冬季极端低温天数，提升年度出行适宜性——尤其在季节性寒冷城市中更为显著。总体而言，这些研究结果构建了一个整合性分析框架，用于理解微出行系统的韧性，并揭示城市骑行者主要依据即时感官环境而非抽象健康指标做出决策。

1. 引言

1.1. 共享单车：可持续城市出行解决方案

城市交通与出行系统正经历一场深刻、迅速且必要的变革。随着全球人口日益向大都市区集中，城市不得不面对日益加剧的温室气体排放、局部空气质量恶化以及长期且严重制约经济发展的交通拥堵等复合危机（Ding 等，2025；Gifford 与 Steg，2005；Mo 等，2018；Wang 与 Zhou，2017）。为应对这些多方面的城市挑战，城市规划者与政策制定者已将向低碳、活跃及共享型交通模式转型列为优先事项（Winkler 等，2023）。在这一转型背景下，共享微出行——尤其是有桩与无桩共享单车系统——已成为现代可持续城市交通网络的基础组成部分。

共享单车系统已发展成为关键的城市基础设施，为数百万城市居民提供了灵活且经济便捷的出行选择（Fishman，2016；Shaheen 等，2010）。这些系统的快速扩展与普及，通过减少乘客行驶里程实现脱碳，直接推动了广泛的气候变化缓解工作（Chapman，2007；Creutzig 与 He，2009）。除全球气候目标外，高密度共享单车网络的局部整合，还能通过转移内燃机产生的人为热量，降低城市地表热岛效应强度（Lee 与 Berkelhammer，2025）。此外，减少车辆尾气排放本身可改善局部空气质量，而鼓励主动出行则能带来连锁性的公共卫生效益，包括增强心血管健康、降低长期使用者的血压，以及从根本上提升城市宜居性（Agarwal 等，2024；Guo 等，2022）。

然而，尽管自行车共享系统带来了显著的社会与环境效益，其韧性仍面临严格的运营审查。与封闭式公共交通或私家车不同，骑行需要直接、不受干预地暴露于周围大气环境中。因此，骑行需求具有高度弹

* 通讯作者：美国伊利诺伊大学芝加哥分校地球与环境科学系，芝加哥，伊利诺伊州，60607。
*电子邮件地址：*jangho.lee@nyu.edu（J. Lee）。

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is highly elastic and sensitive to external meteorological and environmental disruptions (Cortez-Ordonez et al., 2024; Eren and Uz, 2020). Evaluating the long-term viability and spatial resilience of this crucial sustainable transportation mode under the pressures of a changing climate requires a rigorous, multidimensional approach that transcends isolated case studies.

1.2. Literature review

As a fully exposed mode of active transportation, the efficacy of a bike-sharing system in sustaining modal shifts is highly contingent upon the ambient environmental conditions. The decision to cycle is evaluated daily (and often hourly) by the user, requiring a continuous interaction with the physical environment. To synthesize what is presently known and to conclusively justify the research gaps addressed in this manuscript, here we review the triad of environmental determinants governing cycling behavior: meteorological weather, localized air quality, and the macro-level projections of future climate change.

1.2.1. Meteorological stressors: The non-linear dynamics of temperature and precipitation

Weather variables, particularly ambient temperature and precipitation, are recognized in the literature as the most immediate, potent, and highly localized determinants of daily bike-sharing usage (Corcoran et al., 2014; El-Assi et al., 2017; Gong et al., 2024). The relationship between temperature and cycling can be complex, non-linear, and largely dictated by the physiological limits of human thermal comfort. It is widely theorized to follow an inverted U-shape distribution. Warmer temperatures generally encourage cycling adoption and duration up to a specific optimal threshold point. Beyond this threshold, heat stress, which is characterized by excessive sweating, accelerated fatigue, and the physiological dangers of hyperthermia, becomes an overwhelming psychological and physical deterrent (An et al., 2019; Gebhart and Noland, 2014).

One of the most comprehensive syntheses of weather effects is the comparative analysis by Bean et al. (2021). Analyzing nearly 100 million cycling trips across forty cities spanning five distinct global climate zones, the study established that in the vast majority of urban environments, ridership steadily increases alongside temperature until it peaks at approximately 27°C to 28°C, after which usage begins a sharp decline. Furthermore, their multi-city comparison revealed the critical role of regional acclimatization; populations residing in traditionally colder, boreal, or temperate climates exhibit entirely different thermal optimal zones and heat tolerances compared to populations acclimatized to tropical or subtropical zones.

While temperature dictates the overarching, predictable seasonal curve of bike-sharing activity, precipitation acts as an acute, immediate, and nearly universal barrier (Kharaghani et al., 2023; Kim, 2018). The mechanisms by which rain deters cycling are multifaceted: it introduces severe physical discomfort, dramatically degrades road surface traction, impairs both the cyclist's and motorist's visibility, and drastically elevates the perceived and actual risk of traffic collisions. Empirical studies consistently indicate that even marginal accumulations of rainfall can trigger immediate and substantial reductions in network-wide ridership.

1.2.2. Atmospheric stressors: Routine air pollution versus acute visible hazards

In conjunction with the immediate physical sensations of temperature and precipitation, ambient air quality has emerged as a highly critical, yet behaviorally complex, modifier of active travel. Because cycling elevates cardiovascular activity and respiratory rates, cyclists inhale significantly higher volumes of air per minute than sedentary pedestrians or passengers in enclosed, filtered motorized vehicles. This physiological reality forces cyclists to absorb a disproportionately higher burden of ambient urban pollutants (Huang et al., 2024; Raza et al., 2018).

A rapidly expanding body of literature confirms that elevated concentrations of fine particulate matter (PM2.5), ground-level ozone (O3), and carbon monoxide (CO) can act as statistical deterrents to bike-sharing usage (Hong et al., 2022; Liang et al., 2023; Park et al., 2022). However, a critical distinction must be drawn in the literature between measured pollution and perceived pollution. Unlike temperature extremes or heavy rainfall, which are instantly perceptible and demand an instantaneous behavioral response, the insidious health risks associated with routine, baseline urban air pollution (such as standard traffic exhaust) is often invisible and scentless. Consequently, behavioral responses to routine air pollution are heavily mediated by public perception, individual environmental awareness, and the efficacy of public health communication (Kelly et al., 2012; Pascal et al., 2021).

When air pollution crosses the threshold from an invisible metric to an acute, highly visible environmental hazard, the behavioral paradigm shifts drastically. The literature notes that extreme pollution events, most notably the incursion of dense wildfire smoke into urban environments, can degrade visual acuity, induce immediate olfactory discomfort, and create a tangible sensory deterrent. Under these conditions of visible hazard, studies document acute and immediate contractions in active transportation volumes, as the abstract health risk becomes undeniably concrete, prompting users to abandon cycling entirely or seek refuge in enclosed transit modes (Doubleday et al., 2021; Hesseln et al., 2003). Understanding this dichotomy between invisible routine pollution and visible acute hazards is essential for accurately modeling micromobility resilience.

1.2.3. Climate change: Shifting baselines and long-term micromobility resilience

Anthropogenic climate change is fundamentally and irrevocably altering the temporal distribution and intensity of weather patterns globally. Cities are bracing for the intensification and increased frequency of extreme summer heat waves, while simultaneously projecting a systemic contraction in the duration and severity of cold winter seasons. The implications of this shifting climatological baseline for the long-term viability of outdoor, active urban mobility are profound, complex, and highly geographically divergent.

Studies have begun to address this intersection. For instance, the work by Heaney et al. (2019) utilized downscaled climate models specifically for New York City to demonstrate a projection: a warming climate could increase the overall annual volume of bikeshare ridership over the 21st century. This net-positive effect is hypothesized to occur because the anticipated reduction in highly prohibitive, sub-freezing winter days, combined with the temporal expansion of mild “shoulder seasons” (spring and autumn), generates a surplus of cyclable days that mathematically outweighs the ridership losses incurred during the increasingly severe, but temporally concentrated, extreme summer heatwaves.

However, a major limitation of the current literature is the geographic restriction of these projections. It remains insufficiently explored how the climate-driven projections formulated for a specific temperate zone like New York City generalize across a broader spectrum of cities with different baseline climates. Integrating these divergent climate trajectories into a unified analysis is a critical missing link in the macro-level planning of sustainable, climate-resilient mobility infrastructure.

1.3. Research gaps and contributions

Through the synthesis of the aforementioned literature, several pressing research gaps emerge which this study seeks to address.

First, while some studies rely on isolated, single-city case studies (An et al., 2019; Gong et al., 2024; Heaney et al., 2019; Hong et al., 2022; Zheng et al., 2025), the field has also advanced toward multi-city comparative frameworks (Kou and Cai, 2019; Mahajan & Argota Sánchez-Vaquerizo, 2024; Villarrasa-Sapina et al., 2024). However,

性，且对外部气象及环境干扰极为敏感（Cortez-Ordonez 等，2024；Eren 与 Uz，2020）。在气候变化压力下，评估这一关键可持续交通方式的长期可行性与空间韧性，需要一种超越孤立案例研究的严谨、多维分析方法。

1.2. 文献综述

作为一种完全暴露于环境中的主动出行方式，共享单车系统在促进出行方式转型方面的成效，在很大程度上取决于周边环境条件。使用者每日（甚至常以小时为单位）都会对骑行决策进行评估，这要求其与物理环境保持持续互动。为整合现有研究成果，并系统论证本论文所探讨的研究空白，本文综述了影响骑行行为的三大环境决定因素：气象天气条件、局部空气质量以及未来气候变化的宏观预测。

1.2.1. 气象胁迫因子：温度与降水的非线性动力学特征

天气变量，尤其是环境温度与降水量，在文献中被公认为每日共享单车使用量最直接、最强效且高度局部化的决定因素（Corcoran等，2014；El-Assi 等，2017；Gong 等，2024）。温度与骑行行为之间的关联可能呈现复杂、非线性的特征，且主要受人体热舒适度的生理极限制约。学界普遍认为该关系遵循倒U型分布：在特定最优阈值点之前，较高温度通常促进骑行行为的普及与时长增加；一旦超过该阈值，热应激——表现为过度出汗、加速疲劳及高热症的生理风险——便会成为压倒性的心理与生理阻碍（An等，2019；Gebhart与Noland，2014）。

关于天气效应的最全面综述之一，是Bean 等人（2021）进行的对比分析。该研究分析了分布在五个不同全球气候区的四十个城市中近一亿次骑行行程，得出结论：在绝大多数城市环境中，骑行量随温度上升而稳步增加，直至在约27°C至28°C时达到峰值，此后骑行量开始急剧下降。此外，他们的多城市对比研究揭示了区域适应性的重要作用：居住于传统寒冷、寒带或温带气候区的人群，其适宜温度区间与耐热性，与已适应热带或亚热带气候区的人群相比，存在显著差异。

尽管温度决定了共享单车活动总体且可预测的季节性曲线，但降水则成为一种即时、直接且几乎普遍存在的障碍（Kharaghani 等，2023；Kim，2018）。雨水抑制骑行行为的机制是多方面的：它会带来严重的身体不适，显著降低路面抓地力，削弱骑行者与驾驶员的能见度，并大幅增加交通事故的感知风险与实际风险。实证研究一致表明，即使是少量的降雨，也可能导致整个网络范围内的骑行量立即大幅减少。

1.2.2. 大气胁迫因素：日常空气污染与急性可见危害的对比

与温度及降水等即时物理感受相结合，环境空气质量已成为促进主动出行的一项至关重要、但在行为层面却相当复杂的调节因素。由于骑行会增加心血管活动及呼吸频率，骑行者每分钟吸入的空气量显著高于久坐的行人或乘坐封闭式过滤机动车的乘客。这一生理现实迫使骑行者承受不成比例地更高的城市环境污染物负荷（Huang 等，2024；Raza 等，2018）。

大量最新文献证实，细颗粒物（PM2.5）、地面臭氧（O3）和一氧化碳（CO）浓度升高，可能对共享单车使用产生统计学上的抑制作用（Hong 等，2022；Liang 等，2023；Park 等，2022）。然而，文献中必须对实测污染与感知污染作出关键区分。与极端温度或强降雨不同——后者可被即时感知并需要即刻的行为响应——日常基础城市空气污染（如常规交通尾气）所带来的隐蔽健康风险往往不可见、无气味。因此，人们对日常空气污染的应对行为，在很大程度上受公众认知、个人环境意识以及公共卫生传播效果的影响（Kelly 等，2012；Pascal 等，2021）。

当空气污染从一种无形的指标转变为一种急性、高度可见的环境危害时，人们的行为模式会发生剧烈变化。文献指出，极端污染事件——尤其是浓烈的野火烟雾侵入城市环境——会降低视力、引发即时的嗅觉不适，并造成切实可感知的感官障碍。在这些可见危害的条件下，研究证实，主动出行量会急剧且即时地减少，因为抽象的健康风险已变得无可否认地具体化，促使使用者完全放弃骑行，或转向封闭式交通方式（Doubleday et al.，2021；Hesseln et al.，2003）。理解这种无形日常污染与可见急性危害之间的对立关系，对于准确构建微出行韧性模型至关重要。

1.2.3. 气候变化：基准线的偏移与长期微出行韧性

人为气候变化正在从根本上且不可逆转地改变全球天气模式的时间分布与强度。城市正积极应对夏季极端热浪的加剧与频发，同时预计冬季寒冷季节的持续时间与严重程度将呈现系统性缩减。这一气候基准线的变迁，对户外活跃城市出行的长期可持续性所产生的影响，既深刻而复杂，又因地理区位的不同而呈现显著差异。

已有研究开始探讨这一交叉领域。例如，Heaney 等人（2019）的研究采用了专门针对纽约市缩放的气候模型，以展示一种预测：在21世纪，气候变暖可能会增加自行车共享服务的年度总使用量。这种净正向效应被推测源于以下原因：预计极端严寒的冬季天数将减少，加之温和的“过渡季节”（春季与秋季）在时间上的扩展，共同产生了更多可供骑行天数，从数学角度而言，这一数量超过了在日益严重但时间上集中的极端夏季热浪期间自行车使用量的减少。

然而，现有文献的一个主要局限在于这些预测结果的地理范围受限。针对纽约市等特定温带地区制定的气候驱动型预测结果，如何推广至具有不同基础气候特征的更广泛城市群，这一问题仍未得到充分探讨。将这些差异化的气候演变轨迹整合至统一分析框架中，已成为可持续、具气候韧性的交通基础设施宏观规划中一个关键缺失环节。

1.3. 研究空白与贡献

基于对上述文献的综述，本研究识别出若干亟待填补的研究空白，并致力于予以解决。

首先，尽管一些研究依赖于孤立的单城市案例研究（An等，2019；Gong 等，2024；Heaney 等，2019；Hong 等，2022；Zheng 等，2025），但该领域也已朝着多城市比较框架方向发展（Kou与Cai，2019；Mahajan & Argota Sa，Sánchez-Vaquerizo，2024；Villarrasa-

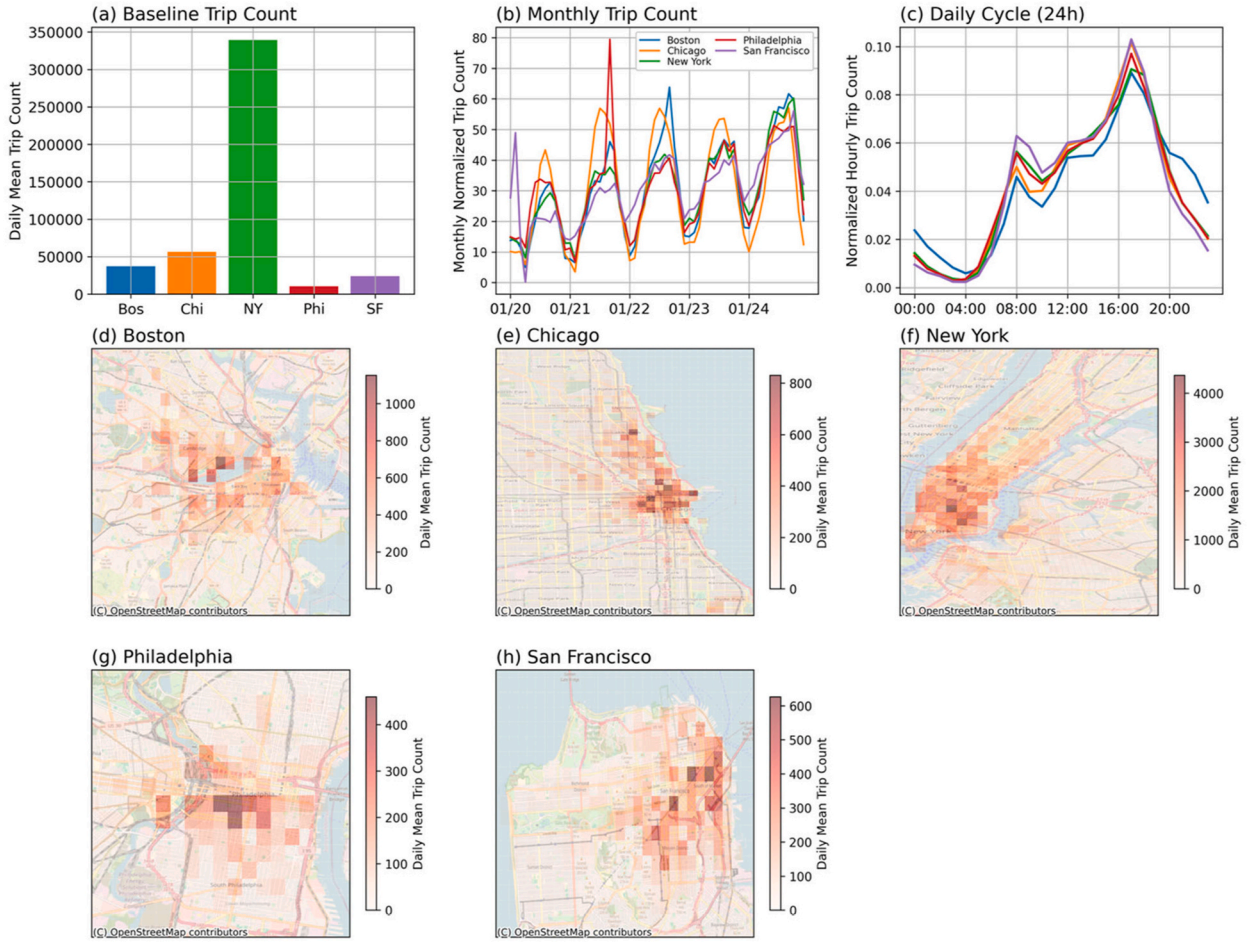


Fig. 1. Spatiotemporal characteristics of bike-sharing usage across five U.S. cities (2020–2024). (a) Daily average trip counts. (b) Normalized monthly ridership. (c) Normalized average 24-h (diurnal) cycle. (d–h) Spatial distribution of mean daily trip counts for Boston, Chicago, New York, Philadelphia, and San Francisco, respectively.

despite this vital progress, a more holistic framework that integrates comprehensive air quality stressors alongside weather and extends these historical sensitivities to project future system resilience under standardized climate change scenarios across contrasting climate zones, is needed.

Second, and perhaps most critically, the spatial dimension of environmental stress on bike-sharing networks must be reconsidered. Static spatial heterogeneity has already been documented in previous studies (Corcoran et al., 2014; Li et al., 2019). Therefore, simply proving that spatial heterogeneity exists is no longer novelty. The underexplored issue is how the spatial concentration of bike-sharing activity is reconfigured under transient environmental stress. When extreme events occur, total ridership may not decline uniformly across the city; instead, statistically significant activity hot spots may persist, disappear, or newly emerge in specific areas. Dense, heat-trapping downtown cores may experience localized network failure, while compensatory cycling activity surges in specific "climatic refuges," such as coastal waterfronts or dense urban park corridors with shade (Cao et al., 2024). Understanding this spatiotemporal reconfiguration is crucial for a complete operational understanding of a network's resilience.

To systematically address these theoretical and methodological gaps, this study provides a comprehensive spatiotemporal assessment of how weather and air quality influence bike-sharing ridership, and how these intricate dynamics will evolve under future climate change scenarios. Deploying an integrated computational framework across five major U. S. cities, this research is driven by three core objectives.

- Temporal Non-linear Analysis across Diverse Climates: Utilizing a flexible statistical model to capture the complex, non-linear responses of ridership to both primary weather variables and diverse air quality metrics.
- Tracking Dynamic Spatial Reconfiguration of Activity Concentration: Moving beyond static spatial heterogeneity by using spatial hot spot analysis to identify where bike-sharing activity hot spots are maintained, lost, or newly formed under environmental stress.
- Future Climate Projections and System Resilience: Projecting these empirically derived historical relationships up to the year 2100 by utilizing high-resolution, bias-corrected climate model projections, providing vital trajectories for urban micromobility resilience in a warming world.

2. Data and method of analysis

2.1. Bike sharing data

This study analyzes publicly available data from bike-sharing systems in five major U.S. cities over a five-year period (January 2020–December 2024). City selection was guided by two main criteria: (1) continuous and reliable operation of the bike-sharing program throughout the study period to ensure temporal completeness, and (2) an average daily ridership exceeding 1000 trips to ensure statistical robustness. The selected cities and their respective systems are Boston (Bluebikes), Chicago (Divvy), New York City (Citi Bike), Philadelphia

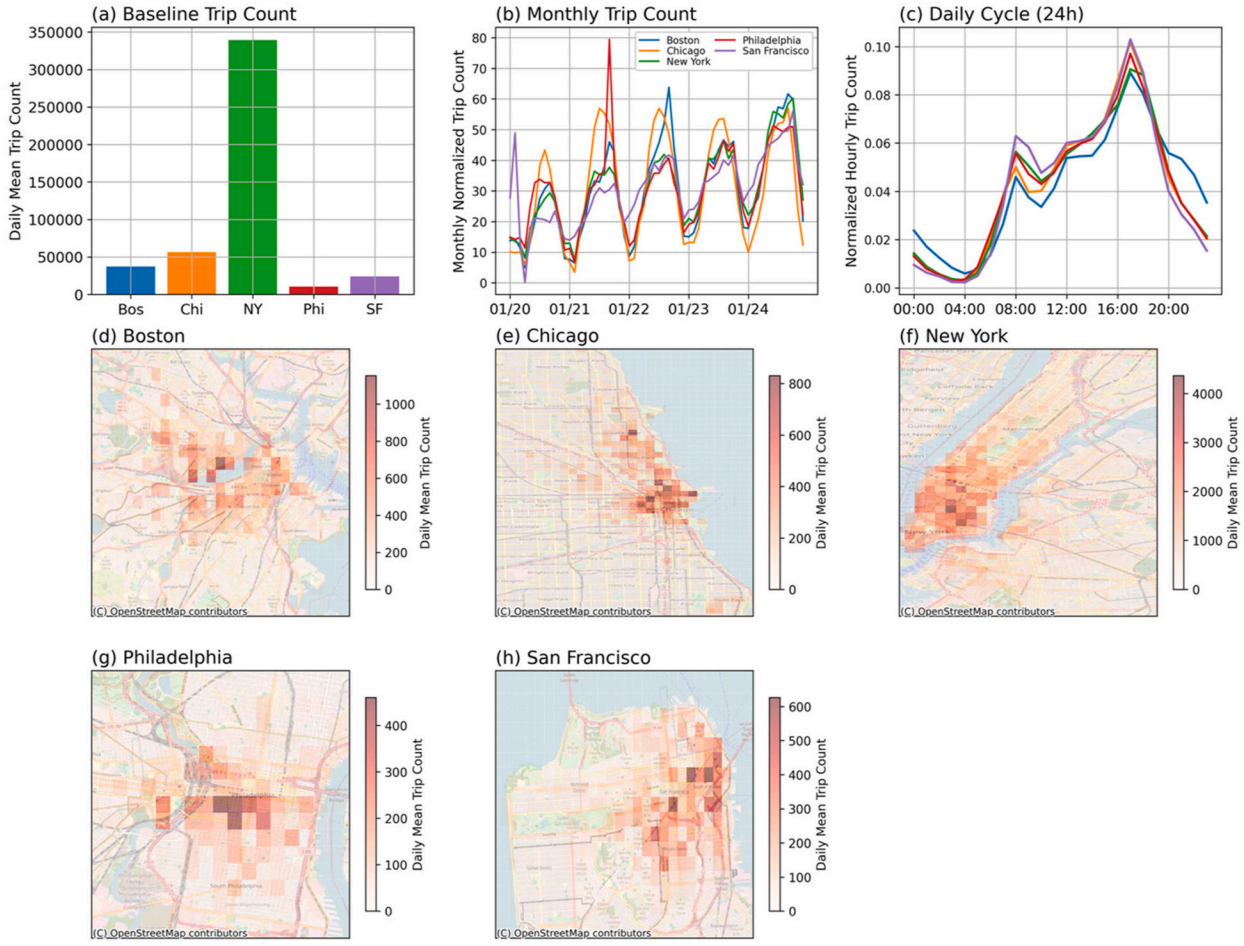


图 1.美国五个城市共享单车使用情况的时空特征（2020–2024 年）。(a)每日平均出行次数；(b)标准化月度骑行量；(c)标准化 24 小时（昼夜）周期平均值；（d–h）波士顿、芝加哥、纽约、费城及旧金山的日均出行次数空间分布图。

Sapina等，2024）。然而，尽管取得了这一重要进展，仍亟需建立一个更全面的框架：该框架需整合全面的空气质量压力源与气象因素，并将这些历史敏感性延伸至标准化气候变化情景下，以评估不同气候区未来系统的韧性。

其次，或许也是最关键的一点，是对共享单车网络所受环境压力的空间维度进行重新考量。静态空间异质性已在既往研究中得到证实（Corcoran 等，2014；Li 等，2019）。因此，仅证明空间异质性的存在已不再具有新颖性。尚未充分探讨的问题是：在短暂环境压力下，共享单车活动的空间分布将如何重新配置。当极端事件发生时，全市范围内的总骑行量未必会均匀下降；相反，在特定区域，具有统计学显著性的活动热点可能持续存在、消失或新现。密集且易积聚热量的市中心核心区可能出现局部网络故障，而在特定“气候避难所”——如沿海滨水区或拥有遮荫的密集城市公园廊道——则可能出现补偿性骑行活动激增（Cao 等，2024）。理解这种时空重构，对于全面把握网络韧性的运行机制至关重要。

为系统性地弥补上述理论与方法论上的空白，本研究对天气与空气质量如何影响共享单车使用量进行了全面的时空评估，并探讨了这些复杂动态关系在未来气候变化情景下将如何演变。本研究在涵盖美国五大主要地区的综合计算框架下开展了相关分析。

针对 S. cities，本研究基于以下三个核心目标展开。

- 跨不同气候条件下的时间非线性分析：采用灵活的统计模型，以捕捉客流量对主要气象变量及多种空气质量指标所呈现的复杂非线性响应。
- 追踪活动浓度的动态空间重构：通过采用空间热点分析，识别在环境压力下共享单车活动热点区域得以维持、消失或新形成的空间分布，从而突破静态空间异质性的研究局限。
- 未来气候预测与系统韧性：利用高分辨率、已校正偏差的气候模型预测结果，将这些基于实证数据推导出的历史关联关系推演至2100年，从而为全球变暖背景下城市微出行系统的韧性提供关键发展路径。

2. 数据与分析方法

2.1. 共享单车数据

本研究分析了美国五大城市共享单车系统在五年期间（2020年1月至2024年12月）的公开可用数据。城市选择基于以下两项主要标准：(1) (1) 在整个研究期间确保共享单车服务持续稳定运行，以保证时间维度的完整性；(2) 每日平均骑行量需超过1000次，以确保统计结果的稳健性。本研究选取的城市及其对应系统分别为：波士顿（Bluebikes）、芝加哥（Divvy）、纽约市（Citi Bike）、费城（Indego）以及旧金山（Bay

(Indego), and San Francisco (Bay Wheels). Cities such as Los Angeles and Washington, D.C. were excluded due to insufficient trip volumes (fewer than 500 trips on most days in Los Angeles) and lack of substantial data coverage (Washington, D.C.).

Each dataset contains detailed trip-level records, including the start and end station coordinates (latitude and longitude) and corresponding timestamps (undocking and docking times). This high-resolution dataset allows for the analysis of spatial usage patterns (by aggregating trip origins and destinations) and their temporal distribution. As illustrated in Fig. 1, the dataset exhibits distinct operational scales and spatio-temporal patterns across the cities. New York City has the highest daily average ridership by a significant margin (approximately 340,000 trips/day), whereas the other four cities have baselines below 100,000 trips/day (Fig. 1a). Looking at the normalized data (divided by daily mean trip count for each city), all systems demonstrate a strong seasonal trend with usage peaking in the summer months (Fig. 1b) and a clear diurnal cycle with bimodal peaks corresponding to morning and evening commuting hours (Fig. 1c). Spatially, bike-sharing activity is heavily concentrated in the dense urban cores of each metropolitan area (Fig. 1d–h).

To preprocess the data, we implemented two filtering steps to remove noise and potential errors. First, spatial outliers were excluded by defining the study area for each city within the 0.5th and 99.5th percentiles of all trip coordinates. This step was implemented to mitigate the influence of GPS positioning errors and extremely long-distance outliers that do not represent typical urban mobility patterns. Second, trips with durations shorter than 60 s were removed, as they are likely to represent false starts or station docking errors. For the main analysis, the filtered trip data were aggregated to a daily temporal resolution and gridded spatially to a resolution of 0.005° in latitude and longitude, corresponding to an approximate grid cell size of 500m × 500m. This spatial resolution was selected to approximate the typical 5-min walking catchment area for bike-sharing stations, balancing the need for fine-grained spatial detail with sufficient data density per cell. To ensure the stability of these results, we conducted comprehensive sensitivity analyses. Testing alternative outlier thresholds (e.g., 1st and 99th percentiles) and alternative spatial grid sizes (e.g., 0.01°) confirmed that the identified spatial hot spot patterns and temporal trends are robust to the choice of preprocessing parameters and spatial units (see Supplementary Section S1 for further details).

2.2. Historical weather and air quality data

Historical daily weather data were sourced from the DayMet V4 dataset (Thornton et al., 2022), which provides gridded climate variables at a 1 km spatial resolution across North America. For this study, we extracted daily total precipitation, specific humidity (sph), and minimum (Tmin) and maximum (Tmax) temperatures. To generate a single representative time series for each metropolitan area, the data from all 1 km grid cells within a 25 km radius of the city center were spatially averaged. The daily mean temperature was calculated as the arithmetic mean of Tmin and Tmax (Thornton et al., 2021), and this, along with total daily precipitation and average daily specific humidity, was averaged across this 25 km radius.

Daily air quality data, including concentrations of carbon monoxide (CO), fine particulate matter (PM2.5), and ground-level ozone (O_3), were obtained from the U.S. Environmental Protection Agency (EPA) Air Quality System. For each city, we used the official city-wide daily summary values, which aggregate observations from multiple standardized monitoring stations to provide a consistent and representative measure of pollutant exposure.

Recognizing that most people are not directly aware of specific pollutant concentrations, we also incorporated the daily Air Quality Index (AQI) for each city, as reported by the EPA (Bishoi et al., 2009; Perlmutt & Cromar, 2019). The AQI offers an integrated and publicly interpretable indicator of overall air quality conditions, facilitating comparison across pollutants and cities. AQI values below 50 are classified as Good, indicating satisfactory air quality. Values between 51 and 100 are considered Moderate, while those from 101 to 150 are categorized as Unhealthy for Sensitive Groups. Values between 151 and 200 are classified as Unhealthy, 201 to 300 as Very Unhealthy, and those exceeding 300 are regarded as Hazardous, posing serious health risks to the entire population.

To test whether ozone and AQI associations partly covary with sunlight, we added DayMet V4 daily shortwave radiation (srad, W/m^2) as a supplementary robustness control. This variable was aligned with the daily ridership and pollution panel and was used only in the additional supplementary models. The baseline historical model remains unchanged. Furthermore, to account for acute wildfire-smoke exposure, we also constructed a daily smoke-exposure indicator using NOAA Hazard Mapping System (HMS) smoke polygons. A city-day was coded as smoke-exposed when an HMS smoke plume intersected the corresponding metropolitan study area. Because wildfire smoke is partly upstream of PM2.5 and AQI, this variable is interpreted as an event-day robustness control rather than as a causal decomposition of particulate pollution. The added-control data construction and HMS smoke-exposure analyses are described in Supplementary Section S2.

Table 1
Global Climate Models (GCMs) and Regional Climate Models (RCMs) from NA-CORDEX used in the projection analysis.

GCM	RCM
CanESM2	CanRCM4
	CRCM5-UQAM
GEMatm-Can	CRCM5-UQAM
GEMatm-MPI	CRCM5-UQAM
GFDL-ESM2M	RegCM4
	WRF
HadGEM2-ES	RegCM4
	WRF
MPI-ESM-LR	RegCM4
	WRF
	CRCM5-UQAM
MPI-ESM-MR	CRCM5-UQAM

2.3. Future weather data

Future climate projections employed in this study were obtained from the North American Coordinated Regional Downscaling Experiment (NA-CORDEX). NA-CORDEX provides dynamically downscaled climate projections generated by coupling multiple Global Climate Models (GCMs) with Regional Climate Models (RCMs), covering the North American domain. Specifically, we utilize projections from 12 distinct GCM-RCM model pairs, as detailed in Table 1, ensuring robust representation of model uncertainty and unforced variability of the climate. GCMs used in this study include CanESM2 (Swart et al., 2019), GEMatm (Šeparović et al., 2013), GFDL-ESM (Dunne et al., 2020), HadGEM2 (Collins et al., 2011), and MPI-ESM (Giorgetta et al., 2013). RCMs used in this study include CanRCM4 (Scinocca et al., 2016), CRCM5-UQAM (Šeparović et al., 2013), RegCM4 (Giorgi et al., 2012), and WRF (Skamarock et al., 2019).

The data utilized in this analysis are available at a daily temporal resolution on the standard NAM-22i grid (approximately 25 km horizontal resolution). To ensure methodological consistency and spatial comparability with our historical observations, the NA-CORDEX projections are bias-corrected and downscaled using the DayMet dataset as the historical baseline period. This downscaling approach aligns the future weather data closely with observed historical climate characteristics. Due to limitations in available datasets, future projections for air quality variables (CO, PM2.5, and ozone) are not included in this analysis.

Wheels）。洛杉矶与华盛顿特区未被纳入研究，原因分别为：骑行量不足（洛杉矶多数日骑行量少于500次）以及数据覆盖范围有限（华盛顿特区）。

每个数据集均包含详细的行程级记录，包括起点与终点站坐标（经纬度）及对应的时间戳（离站与到站时间）。这一高分辨率数据集支持通过分析行程起点与终点的聚合数据，进而探究空间使用模式及其时间分布特征。如图1所示，各城市的数据集呈现出差异化的运营规模与时空分布特征：纽约市的日均骑行量显著领先（约340,000次/日），其余四城市日均骑行量均低于100,000次/日（图1a）。将数据按各城市日均行程数标准化后，所有系统均呈现明显的季节性趋势——使用量在夏季达到峰值（图1b），以及清晰的昼夜周期——早晚通勤时段形成双峰分布（图1c）。从空间分布来看，共享单车使用活动高度集中于各大都市区的密集城市核心区（图1d–h）。

为对数据进行预处理，我们实施了两个过滤步骤以消除噪声及潜在误差。首先，通过将每个城市的研究区域限定在所有行程坐标的第0.5与第99.5百分位区间内，剔除空间异常值。此步骤旨在减弱GPS定位误差及极长距离异常值（这些异常值无法代表典型城市出行模式）的影响。其次，剔除持续时间短于60秒的行程数据，因其可能代表虚假起步或站点对接错误。在主要分析中，将过滤后的行程数据聚合至每日时间分辨率，并在经纬度方向上以0.005 °的网格分辨率进行空间格点化，对应约500米×500米的网格单元尺寸。选择该空间分辨率是为了近似自行车共享站点典型的5分钟步行服务半径，从而在获取精细空间细节与保证每个网格单元内足够数据密度之间取得平衡。为确保研究结果的稳健性，我们开展了全面的敏感性分析：测试不同异常值阈值（如第1与第99百分位）及不同空间网格尺寸（如例如，0.01 °）的结果证实，所识别出的空间热点分布模式与时间趋势对预处理参数及空间单位的选择均具有稳健性（详见补充材料第 S1 节）。

2.2. 历史气象与空气质量数据

历史每日天气数据来源于 DayMet V4 数据集（Thornton 等，2022），该数据集提供了覆盖北美地区、空间分辨率为 1 km 的网格化气候变量。在本研究中，我们提取了每日总降水量、比湿度（sph）以及最低温度（Tmin）与最高温度（Tmax）。为生成每个大都市区的单一代表性时间序列，我们对城市中心 25 km 半径范围内所有 1 km 网格单元的数据进行了空间平均处理。每日平均温度计算为 Tmin 与 Tmax 的算术平均值（Thornton 等，2021），并将该温度值与每日总降水量及每日平均比湿度一同在该 25 km 半径范围内进行平均计算。

每日空气质量数据（包括一氧化碳（CO）、细颗粒物（PM2.5）及地面臭氧（O_3）的浓度）均取自美国环境保护署（EPA）的空气质量监测系统。针对每个城市，我们采用官方发布的全市每日汇总值——该值整合了多个标准化监测站的观测数据，从而提供一致且具有代表性的污染物暴露指标。

考虑到大多数人并不直接了解特定污染物的浓度，我们还纳入了各城市每日空气质量指数（AQI），该数据由美国环保署提供（Bishoi 等，2009；Perlmutt & Cromar，2019）。AQI 提供了一个综合且公众易于理解的空气质量综合指标，有助于不同污染物及城市间的比较。AQI 值低于50被归类为“良好”，表明空气质量令人满意；51至100之间的值被视为“中等”；101至150之间的值则被归类为“对敏感人群不健康”；151至200之间的值被归类为“不健康”；201至300之间的值被归类为“非常不健康”；超过300的值则被归类为“有害”，对全体居民构成严重健康风险。

为检验臭氧与 AQI 的关联是否部分与日照相关，我们加入了DayMet V4每日短波辐射（srad，W/m^2）作为补充稳健性控制变量。该变量与每日客流量及污染面板数据对齐，且仅用于额外的补充模型中。基准历史模型保持不变。此外，为考虑急性野火烟雾暴露的影响，我们还利用NOAA危险制图系统（HMS）的烟雾多边形构建了每日烟雾暴露指标。当HMS烟雾羽流与相应大都市研究区域相交时，将该城市-日标记为烟雾暴露日。由于野火烟雾在部分程度上先于PM2.5与 AQI 出现，此变量被视作事件日的稳健性控制变量，而非颗粒物污染的因果分解指标。新增控制变量的构建方法及HMS烟雾暴露分析详见补充章节S2。

表 1
用于预测分析的 NA- CORDEX 全球气候模型（GCMs）与区域气候模型（RCMs）。

GCM	RCM
CanESM2	CanRCM4
	CRCM5-UQAM
GEMatm-Can	CRCM5-UQAM
GEMatm-MPI	CRCM5-UQAM
GFDL-ESM2M	RegCM4
	WRF
HadGEM2-ES	RegCM4
	WRF
MPI-ESM-LR	RegCM4
	WRF
	CRCM5-UQAM
MPI-ESM-MR	CRCM5-UQAM

2.3. 未来气象数据

本研究中使用的未来气候预测数据均来自北美协调区域降尺度实验（NA- CORDEX）。NA- CORDEX 提供了通过将多个全球气候模型（GCMs）与区域气候模型（RCMs）耦合而生成的动态降尺度气候预测，覆盖北美区域。具体而言，我们使用了 12 对不同的 GCM-RCM 模型的预测结果，如表 1所示，以确保对模型不确定性及气候非强迫变率的稳健表征。本研究中使用的 GCMs 包括 CanESM2（Swart 等，2019），GEMatm（Separovic ‘等，2013）、GFDL - ESM（Dunne 等，2020）、HadGEM2（Collins 等，2011）以及MPI- ESM（Giorgetta 等，2013）。

本研究中使用的 RCMs 包括CanRCM4（Scinocca 等，2016）以及CRCM5- UQAM（Separovic ‘等，2013），RegCM4（Giorgi 等，2012），以及WRF（Skamarock 等，2019）。

本分析所使用的数据以每日时间分辨率提供，采用标准NAM-22i网格（水平分辨率约为25公里）。为确保方法学的一致性及与历史观测数据的空间可比性，NA- CORDEX 投影图均使用DayMet数据集作为历史基准期进行偏差校正及降尺度处理。这种降尺度方法使未来天气数据与历史气候特征高度吻合。由于可用数据集的限制，本分析未包含空气质量变量（CO、PM2.5及臭氧）的未来预测结果。

2.4. Generalized additive models (GAM)

To examine the relationships between shared-bike usage and various environmental factors, we use generalized additive models (GAMs) (Hastie, 2017). GAMs provide a flexible regression framework that allows non-linear responses of the dependent variable to multiple predictors through the use of smooth spline functions. This enables the capture of complex, city-specific behavioral patterns without assuming predefined functional forms.

The baseline model is formulated as:

$$\begin{aligned}\mathrm{Bike}_c = {} & \beta_c + s(\mathrm{temperature}_c) + s(\mathrm{precipitation}_c) + s(\mathrm{CO}_c) + s(\mathrm{PM2.5}_c) \\ & + s(\mathrm{Ozone}_c) + s(\mathrm{time}_c) + \mathrm{is_weekend_holiday}_c + \varepsilon_c\end{aligned}$$

Here, $\mathrm{Bike}_{c,t}$ represents the normalized daily bike usage (divided by the average daily bike usage) time series for city c. β_c is the intercept, and $s(\cdot)$ term represent the smooth spline functions that describe non-linear effects of environmental variables. The binary variable is_weekend_holiday|| is included as a linear term to account for typical day-of-week differences and holidays. The long-term smooth spline function of $s(\mathrm{time}_c)$ is included to control for the seasonal cycle and long-term changes (4 knots per year). In the GAM framework, a knot defines a point along the time variable where separate spline segments join smoothly, thereby determining the flexibility of the fitted curve. This setting allows the spline to capture broad seasonal and long-term trends—such as linear increases in ridership or shifts during the COVID-19 lockdown—while avoiding overfitting short-term fluctuations. ε represents the residual term.

The GAM is fitted separately for each city using restricted maximum likelihood (REML) to optimize the trade-off between model complexity, predictive accuracy, and generalizability. From these fitted models, partial dependence curves are derived to quantify and visualize the independent contribution of each environmental predictor to daily shared-bike usage. Sensitivity tests were conducted on model parameters, including the number of knots per year, and the results remained consistent across configurations (see Supplementary Material Section S3). This stability reflects the inherent smoothing property of the REML-based estimation, which automatically penalizes overfitting and ensures comparable functional forms across cities.

In addition to the pollutant-specific model above, we constructed an alternative GAM using the Air Quality Index (AQI) as a composite indicator. The AQI model replaces individual pollutant terms ($s(\mathrm{CO}_c)$; $s(\mathrm{PM2.5}_c)$; $s(\mathrm{Ozone}_c)$) with a single smooth function $s(\mathrm{AQI}_c)$, while retaining the same terms for temperature, precipitation, time, and weekend/holiday effects:

$$\begin{aligned}\mathrm{Bike}_c^{\mathrm{AQI}} = {} & \beta_c + s(\mathrm{temperature}_c) + s(\mathrm{precipitation}_c) + s(\mathrm{AQI}_c) \\ & + s(\mathrm{time}_c) + \mathrm{is_weekend_holiday}_c + \varepsilon_c\end{aligned}$$

Because AQI is derived from CO, PM2.5, and ozone concentrations, this formulation captures their joint influence through a unified metric rather than as separate predictors. To ensure consistency, the smooth terms for temperature, precipitation, and time in the AQI model were fixed to those estimated in the baseline model, so that the AQI term represents the residual contribution of aggregated air quality to cycling behavior. This two-model structure allows direct comparison between the effects of individual pollutants and their combined perceptual or communicative signal as reflected by AQI.

To identify the most relevant thermal comfort metric for cycling, we tested four different temperature-related variables: standard dry-bulb temperature (DBT), wet-bulb temperature (WBT), Heat Index (HI), and a simplified wet-bulb globe temperature (sWBGT). A comparison of GAM model performance using the R-squared (R^2), Akaike Information Criterion (AIC), and root mean squared error (RMSE) indicated that DBT provided the most explanatory power (see Supplementary Section S4). Consequently, DBT was used for all analyses hereafter.

As a robustness check, we also estimated added-control GAM specifications for both the pollutant-specific and AQI model families by including solar radiation and HMS smoke exposure. These models are reported in Supplementary Section S2 for comparison and are not intended to replace the baseline specification.

2.5. Hot spot analysis

To examine how the spatial concentration of bike-sharing activity responds to extreme environmental conditions, we conducted a hot spot analysis based on the Getis–Ord Gi* statistic. This method evaluates the spatial association among neighboring areas to determine whether high or low usage values are clustered in space beyond what would be expected by random chance. In this framework, a hot spot refers to a location where bike usage is significantly higher than that of its surrounding areas, whereas a cold spot represents a cluster of unusually low activity.

We first constructed two composite (average) maps of daily bike usage for each environmental variable. In this context, the daily bike usage for a specific grid cell is explicitly defined as the total aggregate sum of all trip origins (unlocking events) and trip destinations (locking events) that occurred within the boundaries of that cell on a given day. The Normal Days map was then created by averaging this daily usage data from days when the mean temperature fell between the 25th and 75th percentiles of its distribution. The Extreme Condition Days map was created by averaging days when the environmental variable exceeded its 95th percentile (for high extremes such as heat, rainfall, or air pollution) or dropped below its 5th percentile (for cold extremes). These composite maps allow us to capture stable spatial patterns representative of typical versus extreme conditions.

We defined the reference baseline as all non-event days across the entire study period, rather than using seasonally stratified subsets. This methodological choice was made to capture the absolute deviation of ridership patterns from the system's standard operating state. By comparing extreme events against the annual norm, the analysis reflects the total environmental load—the combined influence of seasonality and the specific stressor—providing a more practical assessment of system resilience and infrastructural stress than a seasonally adjusted metric would offer.

The Getis-Ord Gi* statistic was then calculated for each grid cell i on both maps independently. The statistic for a cell i is given as a Z-score:

$$G_i^* = \frac{\sum_{j=1}^{n} w_{ij}x_j - \bar{X}\sum_{j=1}^{n} w_{ij}}{S\sqrt{\frac{\left[n\sum_{j=1}^{n} w_{ij}^2 - \left(\sum_{j=1}^{n} w_{ij}\right)^2\right]}{n-1}}}$$

Where x_j is the usage value for cell j, w_{ij} is the spatial weight between cell i and j, based on Queen contiguity to include all eight surrounding neighbors. $\bar{X}$ is the mean of all cell values, S is the standard deviation of all cell values, and n is the total number of cells.

Cells with a Z-score greater than 1.645 ($p < 0.1$) were classified as statistically significant hot spots, and cells with negative Z-scores of similar magnitude were classified as cold spots. The analysis was performed separately for the Normal and Extreme composite maps. Comparing these two sets of results allowed us to observe how the “hot spots” of bike activity shift spatially when cities face environmental stressors. Importantly, this method focuses not on day-to-day fluctuations in total trip counts but on the relative reconfiguration of activity within each city. In other words, it helps reveal where people ride more or less under weather stress, not simply how many people ride. Conceptually, this analysis captures whether the spatial “centers of activity” remain stable or migrate when environmental conditions become extreme. Rather than simply showing how total ridership changes, the method reveals whether the concentration of use becomes more dispersed, shifts toward specific neighborhoods, or contracts to

2.4. 广义加性模型（GAM）

为了探究共享单车使用与各种环境因素之间的关系，我们采用了广义加性模型（GAMs）（Hastie，2017）。GAMs 提供了一种灵活的回归框架，通过使用平滑样条函数，允许因变量对多个预测变量呈现非线性响应。这使得我们能够捕捉到城市特有的复杂行为模式，而无需假设预先定义的函数形式。

基准模型表述如下：

$$\mathrm{Bike}_c = \beta_c + s(\text{温度c}) + s(\text{降水量c}) + s(\text{COc}) + s(\text{PM2.5c}) + s(\text{臭氧c}) + s(\text{时间c}) + \text{is weekend holiday}_c + \varepsilon_c$$

此处，$\mathrm{Bike}_{c,t}$ 表示城市 c 的标准化每日骑行使用时长时间序列（即除以每日平均骑行使用时长）。β_c为截距项，而 s（·）项则代表描述环境变量非线性效应的平滑样条函数。二元变量“周末/节假日”被纳入线性项，以考量工作日与节假日的典型差异。引入 s(时间c) 的长期平滑样条函数，用于控制季节性周期及长期趋势变化（每年设置 4 个节点）。在广义样条回归（GAM）框架中，每个节点定义时间变量上一个衔接点，各样条段在此处平滑过渡，从而决定拟合曲线的灵活性。该设定使样条函数既能捕捉广泛的季节性规律与长期趋势——例如骑行量的线性增长或新冠疫情封锁期间的变化——又可避免对短期波动产生过拟合。ε 代表残差项。

针对每个城市，分别使用受限最大似然法（REML）拟合GAM模型，以优化模型复杂度、预测精度及泛化能力之间的权衡。基于这些拟合模型，推导出部分依赖曲线，用以量化并可视化每个环境预测变量对每日共享自行车使用量的独立贡献。对模型参数（包括每年节点数）进行了敏感性测试，结果在不同配置下均保持一致（参见补充材料章节S3）。这种稳定性反映了基于 REML 的估计方法固有的平滑特性，该方法会自动抑制过拟合，并确保各城市间的函数形式具有可比性。

除了上述针对特定污染物的模型外，我们还使用空气质量指数（AQI）作为综合指标，构建了一个替代的广义加性模型（GAM）。该 AQI 模型用单个平滑函数s(AQIc)替代了各个污染物的单独项（s(COc)；s(PM2.5c)；s(Ozonec)），同时保留了温度、降水、时间及周末/节假日效应的相同项：

$$\mathrm{Bike}_c^{AQI} = \beta_c + s(\mathrm{temperature}_c) + s(\mathrm{precipitation}_c) + s(\mathrm{AQI}_c) + s(\text{时间c}) + \text{is-weekend-holiday}_c + \varepsilon_c$$

由于 AQI 源自 CO、PM2.5 和臭氧浓度，这一公式通过一个统一的指标捕捉了这些污染物的共同影响，而非将其作为独立的预测变量。为确保一致性， AQI 模型中温度、降水量及时间的平滑项均固定为基准模型中估计的值，从而使 AQI 项代表综合空气质量对骑行行为的残余贡献。这种双模型结构使得个体污染物的影响与其通过 AQI 反映出的综合感知或传播信号之间的对比成为可能。

为了确定骑行活动中最相关的热舒适度指标，我们测试了四种与温度相关的变量：标准干球温度（DBT）、湿球温度（WBT）、热指数（HI）以及简化湿球全球温度（sWBGT）。通过使用R方（R^2）、赤池信息量准则（AIC）和均方根误差（RMSE）对GAM模型的性能进行比较，结果表明 DBT 具有最强的解释力（参见补充部分S4）。因此，后续所有分析均采用 DBT 作为基准指标。

作为稳健性检验，我们还通过纳入太阳辐射和HMS烟雾暴露，为污染物特定模型族及 AQI 模型族分别估计了添加控制变量的GAM模型设定。这些模型已在补充材料第S2节中列出以供比较，且并非旨在替代基准模型设定。

2.5. 热点分析

为探究共享单车使用活动的空间分布特征在极端环境条件下如何发生变化，我们基于 Getis–Ord Gi* 统计量开展了热点分析。该方法通过评估相邻区域间的空间关联性，判断高或低使用率值是否在空间上呈现超出随机概率预期的聚集现象。在此分析框架中，“热点”指共享单车使用量显著高于周边区域的地点，而“冷点”则指使用量异常偏低的聚集区域。

我们首先为每个环境变量构建了两张每日自行车使用量的综合（平均）分布图。在此定义中，特定网格单元内的每日自行车使用量，被明确定义为该单元边界内某特定日期内所有出行起点（解锁事件）与出行终点（锁定事件）的总量之和。随后，我们将该每日使用量数据中，平均气温处于其分布第25至75百分位区间内的日期数据取平均值，生成“正常天气日”分布图；而对于环境温度变量超过其第95百分位值（针对高温、降雨或空气污染等极端情况）或低于其第5百分位值（针对寒冷极端情况）的日期数据取平均值，生成“极端天气日”分布图。这些综合分布图可帮助我们捕捉到代表典型环境与极端环境下的稳定空间分布特征。

我们将参考基准定义为整个研究期间所有非事件发生日，而非采用按季节分层的子集。这一方法学选择旨在捕捉客流量模式与系统标准运行状态之间的绝对偏差。通过将极端事件与年度基准值进行对比，本分析可反映总体环境负荷——即季节性因素与特定胁迫源共同作用的影响——从而相较于经季节调整的指标，为系统韧性与基础设施负荷提供更具实践价值的评估依据。

随后，分别对两张地图上的每个网格单元 i 独立计算 Getis-Ord Gi* 统计量。对于网格单元 i，该统计量以 Z 分数形式给出：

$$G_i^* = \frac{\sum_{j=1}^{n} w_{ij}x_j - \bar{X}\sum_{j=1}^{n} w_{ij}}{S\sqrt{\frac{\left[n\sum_{j=1}^{n} w_{ij}^2 - \left(\sum_{j=1}^{n} w_{ij}\right)^2\right]}{n-1}}}$$

其中，x_j表示第 j 个单元格的利用值，w_{ij}表示第 i 个单元格与第j 个单元格之间的空间权重，该权重基于皇后邻域原则计算，涵盖所有八个相邻单元格。X 为所有单元格值的平均值，S 为所有单元格值的标准差，n 为单元格总数。

Z值大于1.645（$p < 0.1$）的细胞被归类为具有统计学显著性的“热点”，而Z值为负且绝对值相近的细胞则被归类为“冷点”。该分析分别针对“常规”与“极端”综合图谱进行。通过对比这两组结果，我们可观察当城市面临环境胁迫时，自行车活动“热点”在空间分布上的偏移规律。需特别指出的是，本方法并非聚焦于每日总骑行次数的波动，而是关注各城市内活动分布的相对重构——换言之，该方法旨在揭示在天气胁迫条件下，居民骑行活动发生的区域分布变化，而非单纯统计骑行人数。从概念层面而言，本分析可判断当环境条件趋于极端时，空间“活动中心”是否保持稳定或发生迁移；该方法不仅展示总骑行量的变化，更能揭示使用集中度是呈现更广泛分布、向特定街区集中，还是在

particular corridors under stress. In this way, the hot spot analysis provides an interpretable spatial signature of how urban mobility patterns adapt to adverse environmental conditions.

Unlike the GAM framework where all environmental and non-environmental confounders are mathematically controlled as covariates, the spatial hot spot analysis relies on generating composite average maps. Therefore, to rigorously isolate the spatial signature of each environmental stressor in these maps, we applied strict conditional data filtering. When analyzing the spatial impact of extreme heat, days where precipitation occurred were excluded from the map generation process. This mapping-specific filtering helps ensure that the observed spatial reconfiguration patterns are associated with the target stressor under investigation, while avoiding the dominant confounding suppression of rainfall on spatial activity.

Data from the entire study period were aggregated for the spatial analysis. This approach was chosen to maximize statistical robustness and to identify structurally persistent hot spots that exhibit consistent sensitivity to climate stress over the long term, filtering out transient spatial shifts associated with specific short-term periods.

It is also worth noting that while alternative approaches such as Origin-Destination (OD) flow models or Latent Class Models (LCM) could offer valuable insights into network connectivity and static station typologies, they were not employed in this study. Our primary objective was to pinpoint the dynamic spatial shifts of ridership in response to transient environmental forcing (e.g., heatwaves). For this specific purpose, the density-based Getis-Ord Gi* statistic provides a direct and interpretable measure of localized activity concentration, capturing changes in the spatial configuration of bike-sharing activity without the complexity of route reconstruction.

3. Results and discussion

3.1. Temporal effect of environmental factors

To investigate the city-wide temporal dynamics of bike-sharing, we first modeled the spatially aggregated normalized daily ridership volume against various environmental factors using the GAM framework. The partial dependence curves and their 95% confidence intervals (CIs) reveal distinct non-linear relationships, city-specific sensitivities, and varying degrees of statistical uncertainty (Fig. 2).

Temperature (DBT) exerts the most dominant influence on ridership, driving dependence fluctuations spanning an absolute range of approximately 0.8 (Fig. 2a). Usage reaches a maximum at specific optimal temperatures before declining. For Boston, New York, and Philadelphia, this optimal threshold is consistent, peaking precisely between 21.3°C and 21.7°C. Chicago exhibits an optimal temperature shifted higher to 26.4°C, whereas San Francisco displays a distinctly lower peak at 19.2°C. The narrow 95% CI bands across the core temperature ranges indicate high model confidence in these regional adaptations.

Precipitation consistently demonstrates a steep, negative impact on bike usage (Fig. 2b). Ridership peaks at 0.0 mm/h (no precipitation). The initial dampening effect is substantial and uniform; a light rainfall of just 5 mm/h corresponds to a sharp decrease in the normalized ridership index by 0.10 to 0.13 across all five cities.

Before examining specific pollutants, it is critical to contextualize their absolute magnitudes and statistical uncertainties. While temperature and precipitation show dependence swings of up to 0.8 and 0.4, respectively, all air-quality metrics (AQI, ozone, PM2.5, and CO) are

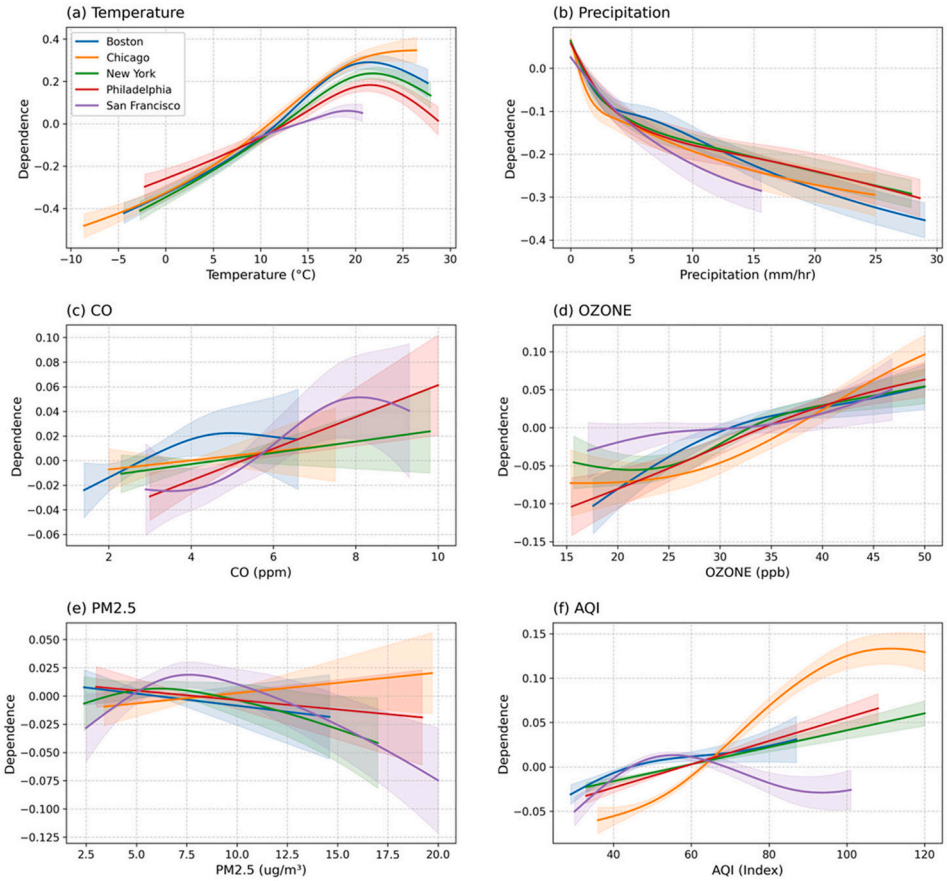


Fig. 2. Partial dependence plots from the city-wide GAMs. The curves illustrate the modeled non-linear relationship between normalized daily ridership and six key environmental variables, holding all other variables constant. The panels show the partial dependence for: (a) Temperature, (b) Precipitation, (c) Carbon Monoxide, (d) Ozone, (e) Fine Particulate Matter (PM2.5), and (f) Air quality index (AQI). 95% confidence intervals (CIs) are shown for all figures as shaded areas.

胁迫条件下收缩至特定廊道。通过这种方式，热点分析可提供一种可解释的空间特征图谱，用以揭示城市出行模式如何适应不利环境条件。

与GAM框架中将所有环境及非环境混杂因素均作为协变量进行数学控制不同，空间热点分析依赖于生成复合平均分布图。因此，为在这些分布图中严格分离各环境胁迫因子的空间特征，我们采用了严格的条件数据过滤方法。在分析极端高温的空间影响时，降水发生日被排除在分布图生成流程之外。这种针对特定制图流程设计的过滤机制，既能确保观测到的空间重构模式与所研究的目标胁迫因子相关联，又可避免降雨对空间活动产生的主要混杂效应干扰分析结果。

本研究将整个研究期间采集的数据进行整合，用于空间分析。采用此方法旨在最大化统计稳健性，并识别出在长期内对气候胁迫始终表现出稳定敏感性的结构性持久热点区域，从而过滤掉与特定短期时段相关的短暂空间位移。

需要特别指出的是，尽管诸如起讫点（OD）流模型或潜在类别模型（LCM）等替代方法可为网络连通性与静态站点类型分析提供有价值的洞见，但本研究并未采用这些方法。本研究的核心目标在于精准捕捉在瞬态环境驱动因素（如热浪）作用下，骑行客流量的动态空间迁移特征。为此，本研究选用基于密度的Getis-Ord Gi*统计量，该指标可直接且易于解读地量化局部活动集中度，从而捕捉共享单车活动空间分布格局的变化，且无需进行路线重构这一复杂步骤。

3. 结果与讨论

3.1. 环境因素的时间效应

为探究共享单车在全市范围内的时间动态特征，我们首先采用 GAM 框架，将空间聚合后的每日标准化骑行量与各类环境因素进行建模分析。偏依赖曲线及其 95% 置信区间（CIs）揭示了明显的非线性关系、城市特有的敏感性，以及不同程度的统计不确定性（图 2）。

温度（DBT）对乘客使用量具有最显著的影响，驱动依赖度波动范围约为0.8（图2a）。使用量在特定最佳温度下达到峰值，随后逐渐下降。对于波士顿、纽约和费城而言，这一最佳温度阈值保持一致，峰值恰好出现在21.3 ˚C至21.7 ˚C之间。芝加哥的最佳温度则偏移至更高的26.4 ˚C，而旧金山的最佳峰值温度明显较低，为19.2 ˚C。在核心温度区间内，狭窄的95%置信区间表明模型对这些区域适应性具有高度置信度。

降水对自行车使用率始终呈现出显著且负面的影响（图 2b）。骑行量在降水量为 0.0 mm/h 时达到峰值。初始抑制效应十分显著且分布均匀：仅 5 mm/h 的轻雨量，即可导致所有五个城市的标准化骑行指数骤降 0.10 至 0.13。

在考察特定污染物之前，至关重要的是将其绝对值及统计不确定性置于适当背景中加以考量。尽管温度和降水量的依赖波动幅度分别可达0.8和0.4，但所有空气质量指标（AQI 、臭氧、PM2.5及CO）的依赖范围

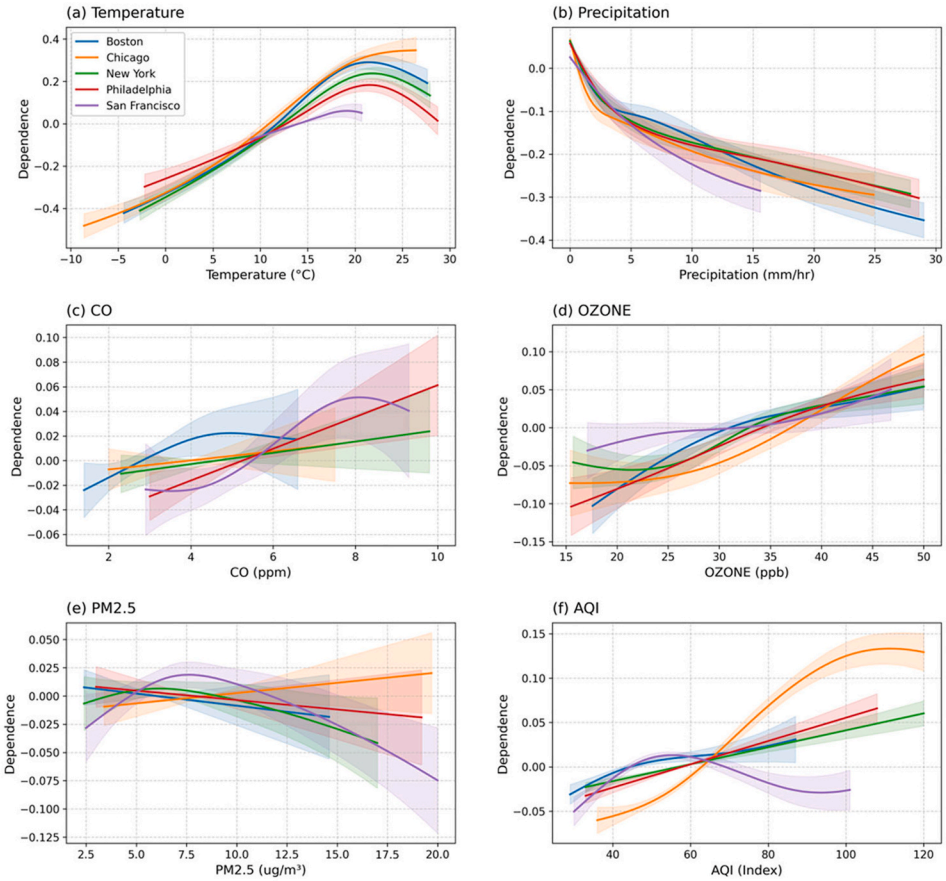


图 2.全市范围 GAM 模型的部分依赖图。这些曲线展示了标准化日客流量与六个关键变量之间建模出的非线性关系。
在保持其他所有变量不变的情况下，考察环境变量对空气质量的影响。各子图分别展示了以下变量的偏依赖关系：(a) 温度，(b) 降水量，(c) 一氧化碳，(d) 臭氧，(e) 细颗粒物（PM2.5），以及 (f) 空气质量指数（AQI）。所有图中均用阴影区域表示 95% 的置信区间（CI）。

constrained within a narrower dependence range of approximately −0.10 to +0.13. The 95% CI shaded regions for pollutants also widen at higher concentration tails, reflecting increased uncertainty due to sparse extreme events. This magnitude difference indicates that weather variables show the largest modeled associations with ridership, whereas routine air-pollution associations are comparatively weak and secondary.

Within this narrower scale, ozone concentration shows a positive baseline association with bike ridership in several cities (Fig. 2d). Because ground-level ozone formation is linked to sunlight and stagnant air, this association may partly reflect covariance with favorable outdoor conditions. The supplementary models that add DayMet shortwave radiation show that the 95th percentile level ozone contrast is attenuated in all five cities after the added controls are included (Supplementary Section S2). We therefore interpret the positive ozone association cautiously as photochemical-weather covariance rather than as a direct beneficial effect of pollution.

Similarly, the partial dependence on AQI displays context-dependent baseline associations (Fig. 2f). In several cities, AQI shows a positive baseline association, but the supplementary models indicate that this association is attenuated after adding solar-radiation and smoke controls (Supplementary Section S2). Because AQI can covary with ozone-related photochemical conditions, the positive AQI response should not be interpreted as evidence that poorer air quality directly increases cycling. We therefore interpret the AQI result as a weak and control-sensitive association rather than a direct behavioral response to pollution.

In stark contrast, the response to PM2.5 and AQI in San Francisco highlights the behavioral tipping point where pollution transitions into a physically perceptible hazard. In San Francisco, PM2.5 shows a clearer negative high-tail response, while the AQI response becomes weaker after adding solar-radiation and HMS smoke controls. The diagnostics in Supplementary Section S2 show that high-tail PM2.5 and AQI days frequently overlap with HMS smoke exposure, including 59.21% of top 5% PM2.5 days and 64.47% of top 5% AQI days. Because wildfire smoke, PM2.5, AQI, and weather conditions are closely linked, we interpret this result as event-day robustness evidence rather than a causal decomposition. The interpretation is therefore that acute high-pollution particulate episodes in San Francisco are associated with ridership reductions, while routine pollution effects remain weaker and city-specific.

To empirically evaluate the counterintuitive air-quality signs, we added threshold-specific robustness tests for AQI, PM2.5, and CO in the Supplementary Material. Across 15 city-pollutant combinations (5 cities × 3 air-quality variables), 8 showed no clear adverse or turning threshold and 5 showed statistically non-significant effects below the identified threshold. Two city-specific PM2.5 cases retained negative below-threshold associations, so we avoid interpreting the threshold pattern as a universal behavioral rule (see Supplementary Section S5). Full added-control model comparisons, solar-radiation attenuation results, HMS smoke diagnostics, and threshold tests are reported in Supplementary Sections S2 and S5.

3.2. Spatial effects of environmental factors

While the city-aggregated GAMs reveal the overall temporal response to environmental factors, the hot spot change analysis (Fig. 3) uncovers the spatial dynamics by showing where these changes in bike-sharing hot spots occur. A prominent pattern observed in Boston, Chicago, and New York is the emergence of a "climatic refuge" effect during periods of heat-related stress. In Boston, under extreme heat and high ozone conditions (Fig. 3, a2, a6), new hot spots (red) emerge along the cooler harbor front. Similarly, Chicago sees the formation of new hot spots along its northern lakefront (Fig. 3, b2), and New York exhibits a subtle but similar shift, with new hot spots appearing on the Hudson River waterfront (Fig. 3, c2, c6). The maps provide strong visual evidence that this is not simply new activity, but a spatial reconfiguration of activity concentration. Critically, as these new waterfront hot spots are activated, existing hot spots within the denser, heat-vulnerable urban cores are simultaneously lost (blue). This pattern supports the suggested trade-off: it indicates that essential, utilitarian cycling in established

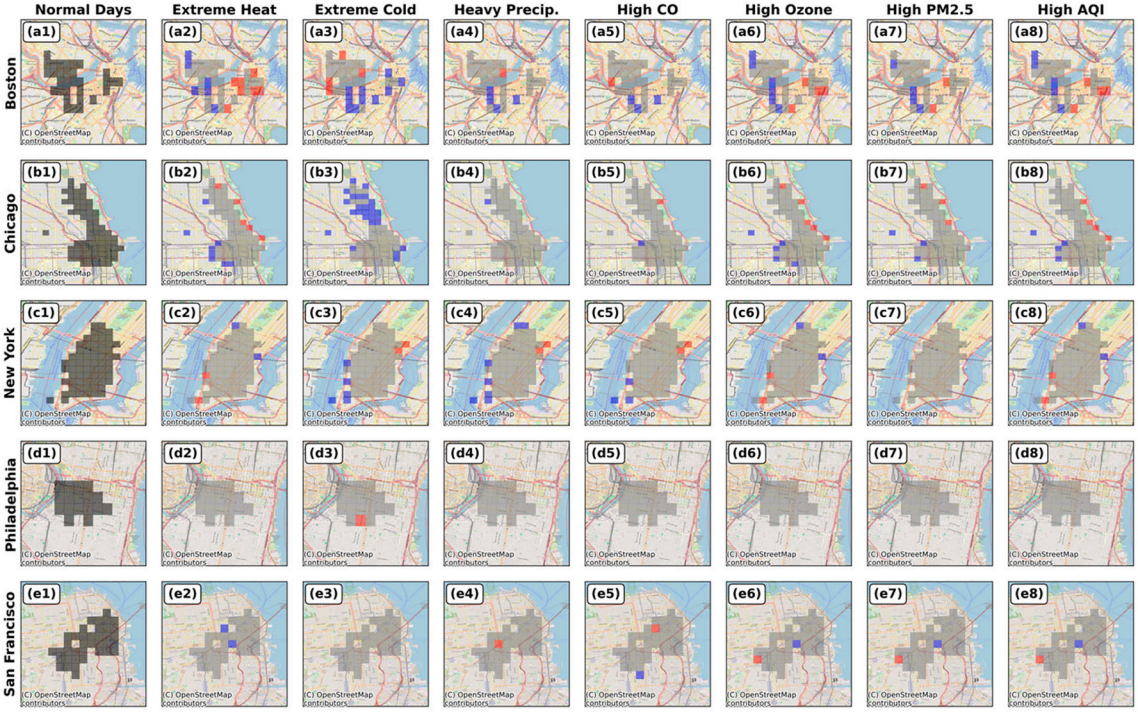


Fig. 3. Hot spot analysis of bike-sharing usage under extreme environmental conditions. Each row corresponds to a city, focusing on areas of concentrated activity. The first column (a1–e1) presents baseline hot spots during Normal Days (dark gray). Subsequent columns (a2–e7) illustrate the spatial patterns observed under each extreme condition, with grid cells classified as Maintained (gray), New (red), or Dispersed (blue) relative to the normal-day baseline.

均较窄，约为_0.10至+0.13。对于各污染物的95%置信区间阴影区域，在较高浓度尾部也会逐渐扩大，这反映了因极端事件稀少而导致的不确定性增加。这一差异表明，天气变量与乘客量之间的建模关联最为显著，而常规空气污染指标与乘客量的关联则相对较弱，仅处于次要地位。

在这一更狭小的尺度内，多个城市中臭氧浓度与自行车骑行量均呈现正向基线关联（图 2d）。由于地表臭氧的形成与日照及空气停滞相关，这种关联可能部分反映了与有利户外条件的共变关系。在引入 DayMet 短波辐射变量的补充模型中，所有五个城市的第 95 百分位臭氧浓度差异均在加入这些控制变量后均出现减弱（补充材料第 S2 节）。因此，我们谨慎地将这种臭氧浓度与骑行量的正向关联解释为光化学-天气条件的共变效应，而非污染物的直接有益作用。

同样地，对 AQI 的部分依赖性也显示出与背景相关的基线关联（图 2f）。在多个城市中， AQI 显示出正向的基线关联，但补充模型表明，在加入太阳辐射和烟雾控制变量后，这种关联会减弱（补充材料第 S2 节）。由于 AQI 可能与臭氧相关的光化学条件共同变化，因此不应将 AQI 的正向响应解读为空气质量恶化会直接增加循环风险的证据。我们因此将 AQI 的结果解释为一种微弱且受控制变量影响的关联，而非对污染的直接行为反应。

相比之下，旧金山对PM2.5和 AQI 的响应凸显了一个行为临界点，即污染逐渐转变为可被物理感知的危害。在旧金山，PM2.5表现出更明显的负向高尾响应，而加入太阳辐射和HMS烟雾控制因素后， AQI 的响应则减弱。补充部分S2中的诊断结果表明，PM2.5和 AQI 的高尾天数常与HMS烟雾暴露时段重叠，包括PM2.5前5%天数中的59.21%以及 AQI 前5%天数中的64.47%。鉴于野火烟雾、PM2.5、 AQI 及天气条件之间存在密切关联，我们将这一结果解读为事件日稳健性的证据，而非因果分解的结果。因此，可以认为旧金山的急性高污染颗粒物事件与乘客量下降相关，而常规污染效应则相对较弱，且具有城市特异性。

为了实证评估这些反直觉的空气质量指标，我们在补充材料中为 AQI 、PM2.5和CO增加了特定阈值的稳健性检验。在15个城市-污染物组合（5个城市×3个空气质量变量）中，有8种组合未显示出明确的负面或转折阈值，另有5种组合在选定阈值以下未表现出统计学显著效应。其中两个城市特有的PM2.5案例仍显示出阈值以下的负相关关系，因此我们避免将阈值模式解读为普适的行为规律（参见补充材料第S5节）。完整的附加控制模型对比、太阳辐射衰减结果、HMS烟雾诊断以及阈值检验均详见补充材料第S2节与第S5节。

3.2. 环境因素的空间效应

虽然城市聚合GAM模型揭示了对环境因素的整体时间响应，但热点变化分析（图3）则通过展示共享单车热点区域的变化发生位置，揭示了其空间动态特征。在波士顿、芝加哥和纽约均观察到一种显著模式：在热应激期间，会出现“气候避难所”效应。在波士顿，极端高温与高臭氧浓度条件下（图3，a2、a6），新的热点区域（红色）沿较凉爽的港口前沿地带涌现；芝加哥则在其北部湖滨区域出现新的热点区域（图3，b2）；纽约亦呈现微妙但类似的迁移趋势——哈德逊河滨水区出现新的热点区域（图3，c2、c6）。这些地图提供了强有力的视觉证据：这并非简单的新活动出现，而是活动集中度的空间重构。关键的是，随着这些新的滨水热点区域被激活，密集且易受高温影响的城市核心区内的现有热点区域（蓝色）却会同步消失。这一模式印证了所提出的权衡关系：在已形成的市中心走廊区域，具有实用功能的必要骑行活动可能会减少

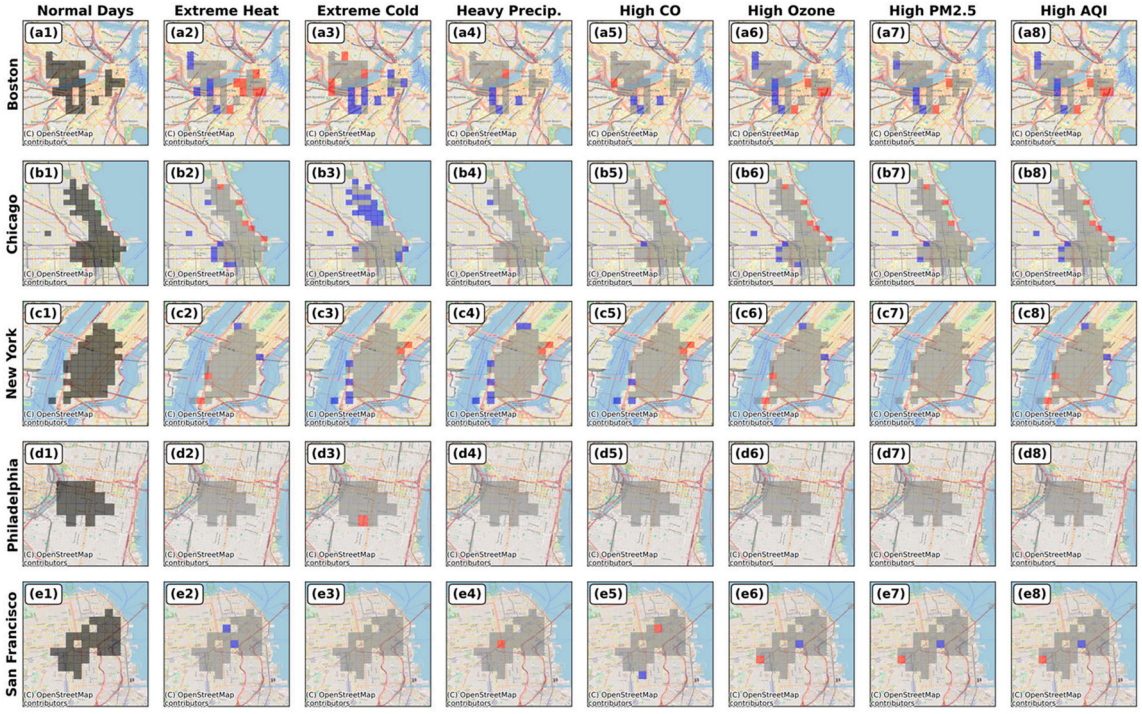


图 3.极端环境条件下共享单车使用情况的热点分析。每行对应一个城市，聚焦于活动密集区域。第一列（a1–e1）展示了正常天气条件下的基准热点分布（深灰色）；后续各列（a2–e7）则分别展示了每种极端条件下观测到的空间分布模式——相对于正常天气基准，网格单元被划分为“维持型”（灰色）、“新增型”（红色）或“分散型”（蓝色）。

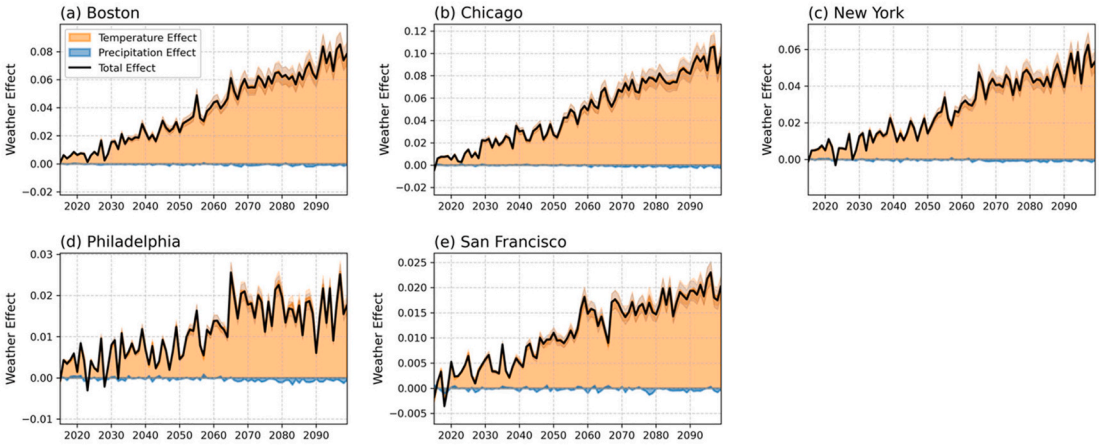


Fig. 4. Projected future temporal trends of the weather-related effect on bike-sharing ridership under the RCP8.5 scenario. Panels show the projected partial effects for (a) Boston, (b) Chicago, (c) New York, (d) Philadelphia, and (e) San Francisco. In each panel, the contributions from projected changes in temperature (orange area) and precipitation (blue area) are shown, along with their total combined effect (black line). The 95% confidence interval is plotted as a shaded band, based on the spread between different NA-CORDEX simulations.

downtown corridors may decline or be disrupted, while recreational or comfort-seeking cycling simultaneously increases or shifts towards areas with more tolerable microclimates, such as those near water bodies.

In contrast, extreme cold triggers a significant hot spot decrease in Chicago (Fig. 3, b3), and the spatial pattern is qualitatively different from the heat-related substitutions. The large, dense usage area in the downtown area visually different. What makes this severe contraction distinct is the lack of compensatory activity; the map is overwhelmingly dominated by lost hot spots, with a near-total absence of new hot spots emerging elsewhere. This suggests that rather than the network adapting or changing, its central hub experiences a net failure under these conditions. The numerous grid cells becoming inactive (dispersed) are not replaced, indicating the downtown core's function as a biking hub is highly vulnerable to winter conditions, possibly as commuters (who might otherwise cycle) switch to enclosed or underground transit options.

The remaining cities, Philadelphia and San Francisco, demonstrate spatial resilience to most environmental stressors, though they achieve this stability through visibly different mechanisms. Philadelphia's compact hot spot cluster is exceptionally stable, showing almost no significant change—minimal red or blue cells—across all extreme conditions (Fig. 3, d1-d7). This lack of dynamics suggests the spatial structure of the core biking activity within this cluster is highly inelastic to environmental factors, continuing largely uninterrupted. San Francisco's hot spot network also maintains its overall structure but shows slightly more dynamic reconfigurations. Rather than being static, its map shows a consistent churn of hotspots (Fig. 3, e1-e7). This pattern is revealed to be a process of local displacement: some cells disperse while new ones emerge nearby. This mechanism allows the network to continuously adapt and re-adjust at a micro-scale in response to various stressors, maintaining its overall integrity without shifting wholesale or failing.

Finally, the High AQI scenario (Fig. 3, a8–e8) reveals spatial patterns that resemble those observed under high ozone conditions. However, the supplementary robustness models show that positive AQI and ozone associations are attenuated after controlling for solar radiation and smoke exposure (Supplementary Section S2). Thus, the spatial similarity is interpreted cautiously as reflecting overlap with warm, sunny, ozone-forming conditions rather than a direct preference for poorer air quality. A sensitivity test regarding the grid size dependency of the hot spot analysis is provided in Supplementary Material Section S1.

3.3. Future projections of weather effects on bike-sharing

To understand the long-term implications of climate change on urban cycling, the fitted GAMs were coupled with future climate projections from the NA-CORDEX ensemble under the RCP8.5 high-emissions scenario. This final stage of the analysis projects how the weather's influence on the suitability of urban cycling is likely to evolve over the 21st century. The results, illustrated in Fig. 4, reveal the potential for a significant and systematic shift in the usage of bike-sharing across all five cities. The most prominent finding is a robust positive trend in the total weather effect (black line) for all metropolitan areas. This suggests that as the century progresses, the net influence of weather will become a significant positive contributor to bike-sharing ridership. This upward trend is overwhelmingly driven by the projected changes in temperature (orange area). The non-linear relationship established in our GAMs (Fig. 2a) indicates that the primary benefit of a warming climate is not the warming of already-hot summers, but rather the reduction of cold, prohibitive days during the winter and the expansion of pleasantly warm conditions in the spring and fall. This expansion effectively lengthens the viable period for comfortable cycling each year. In contrast, the projected changes in precipitation patterns (blue area) have a negative, but negligible long-term impact.

Furthermore, the magnitude of this positive trend is highly heterogeneous and directly linked to the cities' baseline climates. Seasonally cold cities such as Chicago and Boston are projected to experience the largest percentage gains. By 2100, the weather's partial effect in these cities is projected to increase the ridership index by 8 to 10% (which corresponds to +3000 and 5000 daily rides). This amplified effect occurs because these cities are currently the most constrained by cold weather; therefore, they have the most to gain from the reduction of cold days. Conversely, the positive impact is less pronounced in temperate cities like San Francisco (2%, +500 daily rides). Nevertheless, since New York has the highest baseline bike counts (Fig. 1a), it has the largest absolute increase in daily bike counts (+20,000 by year 2100) despite its moderate increase in normalized values (5%).

It is important to emphasize that these projections are based solely on future weather variables (temperature and precipitation), as reliable, downscaled future projections for air quality were not available for this analysis. The projected positive trend in ridership suitability is not a consequence of hotter summers; our GAMs show that extreme heat can be a deterrent. Instead, the primary driver is the significant reduction in

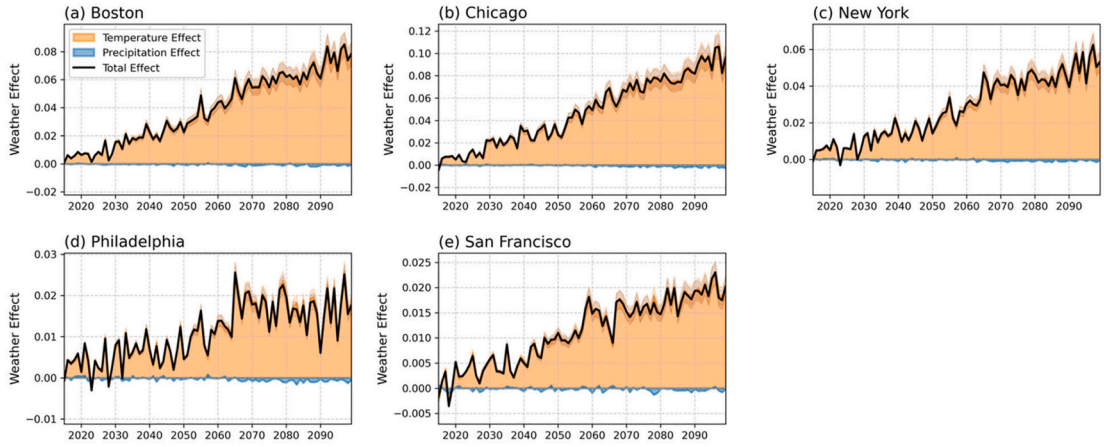


图4.在RCP8.5情景下，天气相关因素对共享单车使用量影响的未来时间趋势预测图。各子图分别展示了(a)波士顿、(b)芝加哥、(c)纽约、(d)费城及(e)旧金山的预测部分影响。每个子图中均显示了温度变化（橙色区域）与降水变化（蓝色区域）的预测贡献，以及两者的综合效应（黑色曲线）。基于不同NA- CORDEX 模拟结果之间的分布范围，95%置信区间以阴影带形式呈现。

或受到干扰；与此同时，休闲性或追求舒适性的骑行活动则会增加，或转向微气候条件更为适宜的区域——例如临近水体的区域。

相比之下，极端低温会导致芝加哥的热点区域显著减少（图 3，b 3），且其空间分布模式在性质上与与高温相关的替代现象截然不同。市中心区域广阔且密集的使用区域在视觉上呈现出明显差异。这种剧烈收缩的独特之处在于缺乏补偿性活动：地图图景中几乎完全由已消失的热点区域主导，而其他地区则几乎未见新热点区域涌现。这表明，在此类条件下，并非整个网络系统发生了适应性调整或演变，而是其核心枢纽出现了净衰退。大量网格单元转为非活跃状态（呈分散分布）且未被替代，说明市中心核心区作为自行车出行枢纽的功能对冬季气候条件极为敏感——这可能是因为通勤者（原本可能选择骑行）转而选择封闭式或地下交通方式。

其余城市——费城与旧金山——对大多数环境胁迫因子均展现出空间韧性，尽管它们实现这种稳定性的机制存在显著差异。费城的紧凑热点集群异常稳定，在所有极端条件下（图3，d1-d7），几乎未出现显著变化——红色或蓝色单元格极少。这种动态性的缺失表明，该集群内核心骑行活动的空间结构对环境因素具有高度刚性，基本持续保持原状。旧金山的热点网络虽也维持整体结构，却呈现出稍具动态性的重构特征：其分布图显示热点区域持续更迭（图3，e1-e7），这种模式实为局部位移过程——部分单元格逐渐消散，而邻近区域则涌现出新单元格。该机制使该网络能够在微观尺度上持续适应并重新调整，以应对各类胁迫因子，既未发生整体性重构，亦未丧失整体完整性。

最后，高 AQI 情景（图3，a8–e8）揭示了与高臭氧条件下观察到的空间分布模式相似的分布特征。然而，补充稳健性模型表明，在控制太阳辐射和烟雾暴露后，正 AQI 与臭氧之间的关联减弱（补充材料第S2节）。因此，这种空间相似性应谨慎解读，它反映的是与温暖、晴朗、易形成臭氧的环境条件相重叠，而非对空气质量较差的直接偏好。关于热点分析中网格尺寸依赖性的敏感性测试，详见补充材料第S1节。

3.3. 未来天气对共享单车运营影响的预测

为了理解气候变化对城市骑行活动的长期影响，我们将拟合的GAM模型与NA- CORDEX 集成模型在RCP8.5高排放情景下的未来气候预测结果相结合。这一分析的最终阶段旨在预测，在21世纪期间，天气对城市骑行适宜性的影响将如何演变。如图4所示，研究结果揭示了所有五个城市中的共享单车使用量可能出现显著且系统性的变化。最突出的发现是，所有大都市区的总天气效应（黑线）均呈现出稳健的上升趋势。这表明，随着世纪的发展，天气的净影响将成为共享单车使用量的重要积极因素。这一上升趋势主要由预测中的温度变化（橙色区域）驱动。我们在GAM模型中建立的非线性关系（图2a）表明，气候变暖的主要益处并非让本已炎热的夏季更加炎热，而是减少冬季寒冷难行的天数，并增加春季宜人温暖天气的时长。以及下降。这一扩张有效延长了每年可舒适骑行期的时长。相比之下，降水模式预计的变化（蓝色区域）虽会产生负面效应，但其长期影响微乎其微。

此外，这一积极趋势的幅度具有高度异质性，且与城市的基线气候条件直接相关。像芝加哥和波士顿这类冬季气候较冷的城市，预计将获得最大比例的增长。到2100年，天气因素对这些城市的骑行指数产生的部分影响预计将使其提升8%至10%（相当于每日骑行量分别增加3000次与5000次）。这种放大效应源于这些城市当前受寒冷天气制约最为严重，因此从减少寒冷天数中获得的收益也最为显著。相反，在旧金山等温带城市，这一积极影响则相对较弱（增幅为2%，每日骑行量增加500次）。尽管如此，由于纽约市的基线自行车使用量最高（图1a），尽管其标准化数值增幅仅为5%，其每日自行车使用量的绝对增幅仍最为显著（到2100年时将增加20,000次）。

需要特别强调的是，这些预测结果仅基于未来气象变量（温度与降水）得出，因为本分析中尚缺乏可靠的、经尺度缩减后的空气质量未来预测数据。乘客适宜性呈现的正向趋势并非源于夏季气温升高；我们的广义加性模型（GAMs）表明，极端高温反而可能构成阻碍因素。相反，其主要驱动因素是相关指标的显著下降。

the number of cold, prohibitive days during the winter and the expansion of optimal riding temperature conditions in the spring and fall.

4. Summary and conclusions

4.1. Summary of the work

This study provides a comprehensive analysis of the spatiotemporal response of bike-sharing systems in five major U.S. cities to multiple environmental stressors, including both weather and air quality. By integrating city-wide temporal non-linear regression (GAM), detailed spatial hot spot analysis (Getis-Ord Gi*), and future climate projections, we aimed to understand the complex and heterogeneous ways in which urban bike sharing mobility systems function under environmental pressure.

Our findings reveal several key insights. First, our temporal analysis confirmed that primary weather variables are dominant drivers of daily ridership. Temperature exhibits a distinct non-linear effect, with an optimal range for cycling between 20°C and 25°C across most cities (Heaney et al., 2019; Zheng et al., 2025). Precipitation acts as a universal deterrent, with even light rainfall causing a sharp and immediate decline in usage (Kharaghani et al., 2023; Kim, 2018). In contrast, air pollutants exhibit a weaker and more complex influence (Hong et al., 2022; Liang et al., 2023; Park et al., 2022). The heterogeneous responses to PM2.5, for example, show a distinct negative trend in San Francisco that is potentially linked to heightened public awareness during wildfire smoke events. Moreover, AQI, which integrates multiple pollutants into a single public-facing metric, displayed mixed associations across cities, reflecting how perceived rather than measured air quality can influence cycling behavior. These findings highlight that the impact of air quality on cycling usage, unlike direct weather effects, is often mediated by local context and public perception. This distinction is important because cycling under heavy air pollution can pose serious health risks (Huang et al., 2024; Raza et al., 2018; Schepers et al., 2015), yet poor air quality is often not directly perceptible in the way temperature or precipitation is. Therefore, the implementation of more active and timely air quality alert systems could play a crucial role in informing and protecting urban cyclists.

Second, our hot spot analysis uncovered distinct city-specific spatial adaptations. We identified a “climatic refuge” effect, where activity shifts towards water bodies during hot conditions, especially in Boston and Chicago. This pattern was most prominent during extreme heat but was also observed, albeit more weakly, during days with high ozone, reinforcing the likelihood that these effects are entangled with temperature. Philadelphia offers a useful warmer-city contrast, with a nearly flat future weather-effect trajectory and no extensive shoreline refuge, suggesting that already warm inland systems may experience both smaller climate-suitability gains and fewer spatial refuge responses under heat stress. We also highlighted the dramatic decrease of Chicago's core hot-spot network and the notable resilience of the networks in Philadelphia and San Francisco, underscoring that the spatial response to environmental stress is geographically specific (Bean et al., 2021; Sarkar et al., 2015; Villarrasa-Sapina et al., 2024).

Finally, by coupling these observed relationships with climate projections, we find that a warming climate is projected to significantly increase the number of days with weather suitable for cycling. This net positive trend is not driven by hotter summers but by the reduction of cold, prohibitive days, which effectively expands the duration of favorable cycling conditions in the spring and fall. The impact is greatest in seasonally cold cities like Chicago and Boston, suggesting a future convergence in the baseline weather suitability for cycling across different North American climates.

4.2. Contributions and implications

The contributions of this work are twofold. Methodologically, we develop an integrated framework that combines GAM analysis with spatial hot spot analysis to evaluate the spatiotemporal resilience of urban bike-sharing systems under multiple environmental stressors. This approach allows for the characterization of complex, city-specific responses that are often obscured in single-variable (Ahn et al., 2022; Noland, 2021), linear and fixed-parameter regression (Kim, 2018; Wessel, 2020) analyses. By capturing non-linear and spatially heterogeneous relationships between environmental factors and shared bike usage, our framework provides a more nuanced understanding of urban mobility dynamics than conventional linear models.

From a practical standpoint, our findings provide critical insights for urban planners and policymakers. The strong non-linear effect of temperature highlights the importance of designing cycling infrastructure and outreach programs that support thermal comfort, such as shaded routes, seasonal bike lanes, and heat mitigation strategies (Cao et al., 2024; Chen et al., 2020; Mateo-Babiano et al., 2016). The discussion on air quality further suggests that simply providing pollution data may not be sufficient to change mobility behavior. The stronger behavioral response observed in San Francisco, where wildfire smoke makes pollution visibly perceptible, indicates that public health messaging must go beyond data dissemination (Doubleday et al., 2021; Hesseln et al., 2003). Communicating the invisible risks associated with pollutants such as PM2.5 or CO in a more tangible and relatable manner may be essential to influencing travel decisions and protecting public health (Kelly et al., 2012; Pascal et al., 2021).

4.3. Limitations and future work

While this study provides a robust multi-city framework, several limitations highlight critical avenues for future research. First, our statistical models do not explicitly incorporate high-resolution spatial data regarding the built environment, such as the presence of protected bike lanes, urban canopy cover, or station network density. As a result, we cannot account for how physical infrastructure might mediate environmental effects. For example, shaded bike lanes may significantly mitigate heat stress, while physically protected lanes may reduce the perceived danger of riding in light rain. Consequently, our findings represent the average environmental sensitivities across the existing, highly heterogeneous infrastructure of each city. It remains unclear whether the observed spatial shifts are driven purely by environmental preferences or are constrained by the pre-existing network of cycling infrastructure.

Relatedly, our spatial analysis relies exclusively on trip origin and destination data rather than the actual routes taken by cyclists. This simplification assumes that environmental conditions at the trip endpoints are representative of the entire journey, potentially overlooking how route choices through diverse microclimates actively mediate a rider's exposure to environmental stressors. Furthermore, to maintain methodological consistency with the 25 km grid of the NA-CORDEX future climate projections, our historical weather and air quality variables were spatially averaged across each metropolitan area. Consequently, our analysis captures the network response to macro-level environmental forcing (such as regional heatwaves or city-wide smoke plumes) but cannot resolve the influence of highly localized microclimates or street level pollution gradients. Finally, our dataset is restricted to public bike-sharing records and does not reflect the behaviors of private bicycle owners, whose travel patterns and demographic profiles may differ. Future research must integrate granular built environment variables, high resolution urban meteorological networks, and route-level inference models to disentangle how urban infrastructure amplifies or buffers the impacts of climate and weather on active mobility.

Methodologically, as an observational study, our framework is designed to capture complex, non-linear spatiotemporal associations rather than to establish strict causal relationships. Although our models rigorously control for concurrent environmental variables and key temporal factors, unobserved confounding variables—including the detailed socioeconomic characteristics of individual riders—may still

冬季寒冷且不宜骑行的天数增多，以及春季与秋季适宜骑行温度条件的扩展。

4. 总结与结论

4.1. 工作总结

本研究对美国五大主要城市的共享单车系统在多种环境压力源（包括天气与空气质量）下的时空响应进行了全面分析。通过整合全市范围的时间非线性回归分析（GAM）、精细化的空间热点分析（Getis-Ord Gi*）以及未来气候预测数据，本研究旨在揭示城市共享单车出行系统在环境压力下呈现出的复杂且异质的运行机制。

我们的研究揭示了若干关键发现。首先，时间分析证实，基础气象变量是每日骑行量的主要驱动因素。温度呈现出明显的非线性效应，在大多数城市中，骑行适宜温度区间为20 °C至25 °C（Heaney 等，2019；Zheng 等，2025）。降水量则普遍抑制骑行行为，即使轻微降雨也会导致骑行量急剧且即时下降（Kharaghani 等，2023；Kim，2018）。相比之下，空气污染物对骑行量的影响较弱且更为复杂（Hong 等，2022；Liang 等，2023；Park 等，2022）。例如，对PM2.5的异质性响应在旧金山呈现出明显的负向趋势，这可能与野火烟雾事件期间公众意识的提升有关。此外，AQI 指标将多种污染物整合为单一公众可感指标，其在各城市间显示出混合关联，反映出感知空气质量而非实测空气质量对骑行行为的影响。这些研究结果表明，与直接天气影响不同，空气质量对骑行使用率的影响往往受到当地环境和公众认知的中介作用。这一区别至关重要，因为在严重空气污染条件下骑行可能带来严重的健康风险（Huang 等，2024；Raza 等，2018；Schepers等，2015），然而，空气质量恶化通常不像温度或降水那样能被直接感知。因此，实施更主动、更及时的空气质量预警系统，可为城市骑行者提供重要信息并发挥关键保护作用。

其次，我们的热点分析揭示了不同城市特有的空间适应性特征。我们发现了“气候避难所”效应，即在高温条件下，活动会向水体转移，这一现象在波士顿和芝加哥尤为明显。这种模式在极端高温期间最为突出，但在臭氧浓度较高的日子里也可见到，尽管强度较弱，这进一步表明这些效应可能与温度密切相关。费城则提供了一个有用的“较暖城市”对比案例，其未来天气效应轨迹近乎平坦，且缺乏广泛的滨水避难所，这表明本就温暖的内陆地区在热应激条件下，可能仅能获得较小的气候适宜性增益，且空间避难所响应也较少。我们还注意到芝加哥核心热点网络的显著缩减，以及费城和旧金山网络所展现出的显著韧性，这强调了环境胁迫下的空间响应具有地理特异性（Bean et al.，2021；Sarkar et al.，2015；Villarrasa-Sapina et al.，2024）。

最后，通过将上述观测结果与气候预测相结合，我们发现：气候变暖预计将显著增加适合骑行天气的天数。这一净正向趋势并非由夏季气温升高驱动，而是源于寒冷且不宜骑行的天数的减少——这实际上延长了春季与秋季适宜骑行的天气时段。该效应在上海、波士顿等季节性寒冷城市中尤为显著，表明未来北美不同气候区域的骑行基础天气适宜性或将趋于趋同。

4.2. 贡献与启示

本研究贡献在于两个方面。在方法论上，我们开发了一个整合框架，将GAM分析与空间热点分析相结合，以评估城市共享单车系统在多种环境压力因素下的时空韧性。这一方法能够刻画城市特有的复杂响应特征，而这些特征在单变量分析（Ahn 等，2022；Noland，2021）、线性及固定参数回归分析（Kim，2018；Wessel，2020）中往往难以捕捉。通过揭示环境因素与共享单车使用量之间的非线性及空间异质性关系，本框架相较于传统线性模型，可提供对城市出行动态更为精细的理解。

从实践角度来看，我们的研究结果为城市规划者和政策制定者提供了关键洞见。温度对骑行行为产生的显著非线性效应，凸显了设计有助于提升热舒适度的骑行基础设施及推广计划的重要性，例如设置遮荫路线、季节性专用自行车道以及实施防暑降温策略（Cao等，2024；Chen等，2020；Mateo-Babiano等，2016）。关于空气质量的讨论进一步表明，仅提供污染数据可能不足以改变出行行为。在野火烟雾使污染状况直观可见的旧金山，观察到的更强行为响应表明，公共卫生宣传必须超越单纯的数据传播（Doubleday等，2021；Hesseln等，2003）。以更具象化且易于理解的方式，向公众传达PM2.5或CO等污染物相关的无形风险，对于引导出行决策及守护公众健康至关重要（Kelly 等，2012；Pascal 等，2021）。

4.3. 局限性与未来研究方向

尽管本研究构建了稳健的多城市研究框架，但若干局限性仍为未来研究指明了关键方向。首先，我们的统计模型尚未明确纳入与建成环境相关的高分辨率空间数据——例如专用自行车道的设置、城市冠层覆盖度或站点网络密度——因此，我们无法评估物理基础设施如何中介环境效应。例如，遮荫式自行车道可显著缓解热应激，而物理防护型自行车道则可能降低小雨骑行时的感知风险。由此可见，本研究结果反映的是各城市现有高度异质性基础设施环境敏感性的平均水平。目前尚不明确，所观察到的空间分布变化究竟纯粹源于环境偏好，还是受既有自行车基础设施网络的制约。

与此相关的是，我们的空间分析仅依赖于出行起点与终点数据，而非骑行者实际选择的路线。这一简化假设认为，出行起点与终点的环境条件能够代表整个旅程的环境状况，但可能忽略了骑行者在不同微气候环境中选择路线时，这些选择如何主动影响其暴露于环境胁迫因素的程度。此外，为了与 NA- CORDEX 未来气候预测中使用的 25 公里网格保持一致的方法论，我们将历史天气及空气质量变量在每个大都市区内进行空间平均。因此，我们的分析能够捕捉网络对宏观环境驱动因素（如区域性热浪或全市范围的烟雾羽流）的响应，但无法解析高度局部化的微气候或街道层面的污染梯度所产生的影响。最后，我们的数据集仅限于公共共享单车记录，未能反映私人自行车使用者的行为特征，后者的出行模式及人口特征可能存在差异。未来研究需整合细粒度的建成环境变量、高分辨率城市气象监测网络以及路线级推理模型，以厘清城市基础设施如何放大或缓冲气候与天气对主动出行行为的影响。

在研究方法上，作为一项观察性研究，本研究框架旨在捕捉复杂且非线性的时空关联，而非确立严格的因果关系。尽管我们的模型已严格控制同期环境变量及关键时间因素，但未被观测到的混杂变量——包括个体骑行者的详细社会经济特征——仍可能存在。此外，在整个研究期

exist. Additionally, the spatial aggregation of data across the entire study period, while effective for identifying structural vulnerabilities, may obscure transient compositional shifts that occurred during the COVID-19 pandemic (e.g., the relocation of activity from commercial hubs to residential areas). Future studies should leverage advanced causal inference designs to definitively isolate the causal effects of environmental and socioeconomic factors on micromobility adoption.

Finally, our long-term climate projections assume that the current sensitivity of ridership to meteorological variables will remain static over time, and they are confined exclusively to primary weather variables due to the current limitations in highly downscaled future atmospheric chemistry models. As such, our projections do not account for future localized changes in air quality. This is a vital limitation because the environmental benefits of a warming climate (e.g., fewer freezing winter days) could be severely counteracted by climate-driven air quality degradation, such as spikes in PM2.5 from increased wildfire frequency. Furthermore, this static approach does not account for potential societal adaptations, such as the widespread adoption of e-bikes, extensive infrastructure expansion, or human behavioral acclimatization to higher temperatures. Therefore, our projections should be interpreted as a baseline assessment of climate-driven thermal pressure on the current system configuration, rather than a precise forecast of future demand. Incorporating these air quality risks and adaptive dynamics, alongside finer-grained socioeconomic indicators, will be essential for developing a comprehensive understanding of urban micromobility resilience in a warming world.

CRediT authorship contribution statement

Jangho Lee: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Max Berkelhammer:** Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

All data used in the article is publicly available.

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间对数据进行空间聚合处理，虽有助于识别结构性脆弱性，却可能掩盖新冠疫情暴发期间出现的短暂性构成变化（例如活动从商业核心区向居住区的转移）。未来研究应采用先进的因果推断设计，以明确分离环境及社会经济因素对微出行采纳行为的因果效应。

最后，我们的长期气候预测基于以下假设：当前客流量对气象变量的敏感性将随时间保持恒定；且由于当前高分辨率未来大气化学模型的技术局限，预测仅涵盖主要气象变量。因此，本预测未考虑未来局部地区空气质量的变化。这一局限至关重要，因为气候变暖带来的环境效益（例如冬季结冰天数减少）可能被气候驱动下的空气质量恶化（如野火频发导致的PM2.5浓度骤升）严重抵消。此外，这种静态预测方法未纳入潜在的社会适应性措施，例如电动自行车的广泛普及、基础设施的大规模扩建，或人类对高温环境的行为适应。故本预测应被视为对当前系统配置下气候驱动型热压力影响的基准评估，而非对未来需求量的精确预测。将上述空气质量风险与适应性动态因素，以及更精细的社会经济指标纳入考量，对于在气候变暖背景下全面理解城市微出行系统的韧性至关重要。

CRediT 作者贡献声明

Jangho Lee：概念化、数据管理、形式分析、调研、方法论制定、项目管理、资源获取、软件应用、监督指导、验证、可视化、初稿撰写、文稿审阅与编辑。**Max Berkelhammer：**资金筹措、监督指导、文稿审阅与编辑。

利益冲突声明

作者声明，其不存在已知的竞争性经济利益或个人关系，这些利益或关系可能对本研究结果产生不良影响。

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附录 A：补充数据

本文的补充数据可在以下网址在线获取：https://doi.org/10.1016/j.jenvman.2026.130431。

数据可用性

本文中使用的所有数据均公开可用。

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